

# Peritoneal Dialysis

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## CHAPTER OUTLINE

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## KEY POINTS

- Contemporary PD prescriptions emphasize goal-directed therapy, taking into account lifestyle factors, residual kidney function, and peritoneal membrane characteristics, with incremental PD offering a gradual approach to dialysis initiation while maintaining comparable outcomes to conventional prescriptions.
- The three-pore model of peritoneal solute transport provides a framework for our understanding of movement of water and solutes across the peritoneal membrane. Genetic studies have highlighted the variability in aquaporin-1 expression in the peritoneal membrane and its vital role in peritoneal water removal. Long-term membrane alterations, including fibrosis and vasculopathy, may significantly limit the long-term viability of PD therapy.
- Updated International Society for Peritoneal Dialysis guidelines emphasize early broad-spectrum antibiotic initiation for suspected peritonitis and highlight the important and key role of quality improvement initiatives and systematic measuring of PD-related infection rates while highlighting key peritonitis preventative strategies.
- Advances in PD include the more widespread availability of assisted PD programs and the use of remote patient monitoring. Global disparities in the availability of dialysis persist, and efforts to increase PD utilization may be particularly advantageous in resource-limited settings, but several obstacles need to be overcome.

## PERITONEAL DIALYSIS—HISTORY AND EPIDEMIOLOGY

In 1877, Wegner conducted pioneering experiments in animal models, exploring the effects of intraperitoneal fluid administration.<sup>1,2</sup> Wegner observed that introducing a concentrated solution of sugar could lead to an increased amount of fluid in the abdominal cavity. This marked the first recorded instance of utilizing the peritoneum for ultrafiltration. In 1923, George Ganter reported the application of peritoneal dialysis (PD) for uremia treatment in animal models after ligation of their ureters, demonstrating the potential of PD in correcting biochemical abnormalities. Concurrently, Tracy Jackson Putnam at Johns Hopkins Hospital conducted parallel investigations, emphasizing the

significance of equilibrium between blood and peritoneal fluid during fluid dwells in the peritoneal cavity.<sup>3</sup>

The clinical application of PD remained dormant until 1946, when J. Frank, A. M. Seligman, and J. Fine successfully employed “peritoneal irrigation” in treating severe acute kidney injury. By 1948, they had treated 18 patients, achieving survival in 4 cases. The clinical application was further bolstered in the 1950s following the introduction of the chronically indwelling Tenckhoff PD catheter.<sup>4-6</sup> Subsequent decades witnessed pivotal advancements including the development of automated PD machines, the refinement of dialysate solutions, and solution storage into plastic containers that shifted its delivery from an institutional-based to a home-based therapy.<sup>1,7</sup> These innovations enhanced the therapeutic efficacy of PD and facilitated its broader acceptance and implementation,

markedly improving patient outcomes. Presently, PD is an integral component of kidney replacement therapy, representing up to 11% of global dialysis treatments serving as a viable alternative to hemodialysis (HD).<sup>8-11</sup>

From a clinical and economic perspective, the benefits of PD are numerous. It emerges as a more economically prudent choice, particularly in higher-income countries.<sup>12-15</sup> Additionally, PD is a continuous therapy, allowing for gradual fluid removal, and mitigating severe hemodynamic fluctuations seen in intermittent HD. Observational studies also suggest better preservation of residual kidney function in patients receiving PD compared with those receiving it in center-based HD.<sup>8</sup> Furthermore, a 10-year-long investigation from Canadian registry data demonstrated that cost associated with PD was about 20% less than that of HD, as measured using quality-adjusted life-years.<sup>12</sup>

Randomized controlled trial analysis of survival outcomes between PD and HD was attempted, but failure of randomization occurred due to the strong preferences potential study subjects had in choosing one modality over another after receiving education.<sup>16</sup> Therefore the majority of evidence stems from large observational studies demonstrating superior survival for patients treated with PD within the first 2 years, subsequently surpassed by HD in the following 2 years.<sup>17,18</sup> However, longitudinal data reveal a convergence of survival rates between PD and HD, with these rates being modulated by variables such as patient age, presence of diabetes, and size of the treatment facility. Findings suggest that the early disparities in survival might be attributed to confounding factors including a higher acuity population initiating in-center HD. When adjusting for these factors, early survival rates for PD and HD appear equivalent.<sup>19</sup> Although long-term survival outcomes vary on the basis of study methodology and specific patient demographics, larger registry studies often generalize findings, lack robust detailed adjustments, and may potentially mask subtle differences in key subgroups.

PD is currently available in approximately 81% of countries, yet its utilization rates show vast disparities, ranging from very low to very high rates across different regions.<sup>8</sup> The higher PD prevalences are prominently seen in China, the United States, Mexico, Thailand, and Scandinavian countries.<sup>20</sup> In stark contrast, its utilization is noticeably limited in economically disadvantaged nations, with the gap being most pronounced on the African continent. Currently, 20 countries in Africa lack the capability to provide PD.<sup>8,20</sup> Several factors contribute to this significant disparity. The high cost of locally manufacturing dialysate solutions presents a major financial hurdle, and when local production is not viable, the prohibitive costs of importing these solutions further strain limited health care budgets.<sup>8,21</sup> Additionally, there is a pronounced shortage of health care professionals skilled in PD, and resources to train such professionals are also limited.<sup>10,21</sup> Despite the aforementioned challenges in low-resource settings, there have been emerging initiatives to address these disparities. The Saving Young Lives (SYL) program, initiated in 2012, represents a promising initiative aimed at implementing sustainable acute PD programs (for treatment of acute kidney injury) in resource-limited settings.<sup>22</sup> By demonstrating success and providing a scalable model, the SYL program holds the potential to extend life-saving treatments globally, particularly in areas most in need. It is possible that access to PD for maintenance dialysis can

be expanded through this program. Global challenges are discussed further in [Chapter 73](#).

However, as with any medical treatment, PD comes with its set of challenges. Historical inconsistencies in PD outcome reporting due to divergent terminologies and varied clinical practices highlighted an imperative for universal guidelines and standardization. In response, initiatives like the Standardized Outcomes in Nephrology in PD (SONG-PD) were introduced.<sup>23,24</sup> The mission of SONG-PD centers around the engagement of various stakeholders, from patients, caregivers, medical professionals, and researchers through a series of collaborative efforts to uncover the most significant outcomes for patients and all kidney care stakeholders.<sup>23,24</sup> Another initiative to promote effective international collaborative research in PD was the launch of the Peritoneal Dialysis Outcomes and Practice Patterns Study (PDOPPS). The main goal of PDOPPS is to pinpoint modifiable practices that lead to better patient and PD technique survival.<sup>25</sup> Working alongside the International Society for Peritoneal Dialysis (ISPD), PDOPPS aims to harmonize data definitions related to PD and establish a platform for global clinical research in PD. As a result, these initiatives have fostered uniformity in outcome definitions and meticulous reporting but also championed best practices in PD, underscoring the potential and the necessity for continuous refinement in the field.

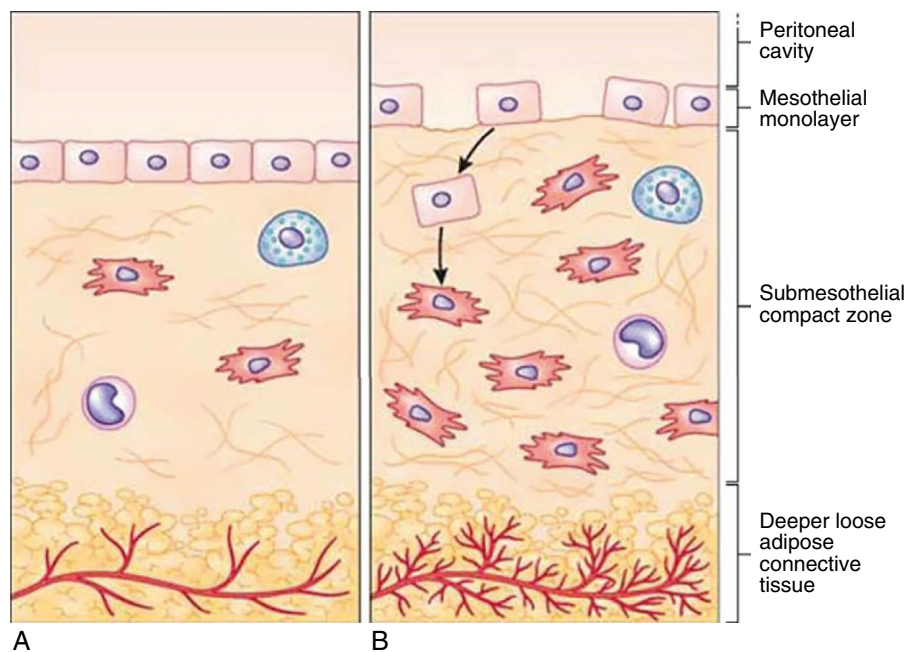
The median time on PD for patients ranges from 1.7 to 3.2 years and is influenced by several factors. First, structural changes and fibrosis to the peritoneal membrane over time lead to decreased efficiency for solute and water removal (see “[Longitudinal Changes in Peritoneal Membrane](#)” later). Second, patients may develop infectious complications such as peritonitis, and non-infectious complications such as PD fluid leaks may shorten time on PD therapy. Finally, patient preferences or changes in their medical condition may warrant a transfer to HD.

## PERITONEAL MEMBRANE ANATOMY AND PHYSIOLOGY

### PERITONEAL MEMBRANE ANATOMY

The peritoneal cavity is lined by a monolayer of mesothelial cells, beneath which lies the submesothelial compact zone.<sup>26-29</sup> This zone houses the interstitium, a gel-like matrix populated with fibroblasts, mast cells, collagen, and various extracellular matrix components. The third layer is the vascular layer, consisting of a capillary endothelium network, an endothelial basement membrane, and a capillary fluid film that overlays the endothelium<sup>29</sup> ([Fig. 63.1](#)).

In an average-sized adult, the surface area of the peritoneal membrane typically ranges between 1.6 and 2.0 m<sup>2</sup>.<sup>28</sup> Microvilli, present on the mesothelial cells, amplify this area by almost 20 times, maximizing membrane dialytic capacity and effectively increasing the functional surface area to approximately 40 m<sup>2</sup>.<sup>30,31</sup> Mesothelial cells, replete with tight junctions, gap junctions, and desmosomes, maintain the structural integrity of the peritoneum and also play an essential role in cellular communication.<sup>31,32</sup> Furthermore, a plethora of glucose transporters and aquaporin channels embedded within the peritoneal microvasculature respond dynamically to both osmotic and nonosmotic stimuli.<sup>29,33,34</sup>



**Fig. 63.1** Changes in the peritoneal membrane over time: initially (A), the peritoneal membrane baseline architecture. After ongoing treatment with dialysis (B), the mesothelial monolayer undergoes changes, leading to an epithelial-to-mesenchymal transition of the cells toward a myfibroblastic morphology. The submesothelial compact zone thickens due to increased extracellular matrix deposition. Deeper in the adipose-rich connective tissue, the vascular network is subject to neovascularization and escalating vasculopathy, marked by hyalinization of arterioles and venules, luminal enlargement, and obliteration. (From Nessim SJ, Perl J, Bargman JM. The renin-angiotensin-aldosterone system in peritoneal dialysis: is what is good for the kidney also good for the peritoneum? *Kidney Int.* 2010;78(1):23–28.)

## SOLUTE AND WATER TRANSPORT

Central to the understanding of PD physiology is the concept of the “effective peritoneal surface area.”<sup>35,36</sup> The effective peritoneal surface area takes into account the perfused capillaries, hence providing a more detailed insight into solute and fluid exchanges.<sup>36</sup> The density and surface proximity of capillaries are key factors that determine the speed of solute transport. This becomes even more relevant when considering pathologic conditions that can modulate this effective area. For example, during peritonitis, the peritoneum undergoes an inflammatory response, which consequently leads to capillary vasodilation. This inflammatory process increases the peritoneal blood flow rate and also expands the effective peritoneal surface area, thereby boosting the diffusion of small solutes while increasing peritoneal protein losses.<sup>37,38</sup> Additionally, the inflammatory response can lead to more rapid absorption of dextrose across the peritoneal membrane, leading to a rapid dissipation of the osmotic gradient, ultimately reducing peritoneal ultrafiltration (UF).<sup>38,39</sup>

At the heart of our current understanding of fluid and solute transport across the peritoneal membrane is the “three-pore model,” a theoretical framework that classifies peritoneal membrane transport of solutes and water across three distinct pore sizes.<sup>40</sup> Small pores, approximately 5 nm in radius, are the most abundant and are responsible for the majority of solute removal during PD. It is believed that these small pores account for more than 95% of the total solute removal, making them central to the dialysis process.<sup>29</sup> Another important pore type with diameters typically under 0.4 nm are aquaporin-1 (AQP1) channels, sometimes referred to as ultras-small pores. The appearance of water-only channels was postulated on the basis of the discovery that instillation of glucose-based PD solution led to an initial

drop in peritoneal sodium concentration, suggesting water-only movement.<sup>34,41</sup> Only later were these discovered to be AQP-1 channels in the peritoneal membrane.<sup>40</sup> Aquaporin channels are crucial for glucose-related osmotically mediated water transport across the peritoneal membrane.<sup>34</sup> Additionally, AQP1 channels contribute to about 50% of the total UF volume in PD.<sup>42,43</sup> Morelle and colleagues<sup>44</sup> highlighted in a landmark study that the functionality of these channels can vary among individuals owing to genetic variations. The analysis from their study revealed that genetic variation related to the expression AQP1 can directly affect peritoneal ultrafiltration. Certain genotype variants associated with diminished AQP-1 expression and lower UF rates were also associated with higher risks of death or transition to HD.

Osmotic conductance, another vital concept in the realm of PD, pertains to the ability of the peritoneal membrane to allow water passage. Contrary to some prevalent misconceptions, osmotic conductance is not directly related to peritoneal blood flow. Instead, factors like density of AQP1 and the aforementioned small pores significantly influence osmotic conductance.<sup>29,45</sup> Additionally, peritoneal fibrosis can reduce conductance, which in turn decreases UF in response to osmotically active solutions.<sup>33</sup> The intricate interplay of osmotic conductance with the unique properties of the peritoneal membrane underscores the complex nature of fluid dynamics in PD.

Beyond the three-pore model, transport across the peritoneal capillaries is heavily dependent on structural properties of the solute. For example, the movement of solutes with low to medium molecular weights is predominantly determined by their molecular dimensions. Sieving involves the selective retention of solutes across the peritoneal membrane, influenced by their varying sieving coefficients. These coefficients, indicative of solute permeability, are

determined by properties including molecular weight and charge.<sup>46</sup> A higher sieving coefficient means greater permeability; for example, urea, with its low molecular weight of 60 Da, has a high coefficient, facilitating rapid equilibration across the membrane.<sup>47</sup> For larger molecules, transport is dependent on additional factors such as the inherent permeability of the peritoneal membrane. Contrary to prevalent assumptions, charge has not been shown to play a significant role in transport of larger molecules.<sup>48</sup>

In contemporary clinical practice, the 24-hour clearance and 4-hour dialysate/plasma (D/P) ratio are now commonly used for evaluating the removal of small solutes like creatinine ( $D/P_{\text{creatinine}}$ ) and urea ( $D/P_{\text{urea}}$ ).<sup>49</sup> These metrics are more convenient and also may guide day-to-day clinical decision making. It is important to underscore that the primary mechanism for the elimination of low-molecular-weight solutes in PD is diffusive transport, with convective transport also playing a role. This understanding is critical for clinicians as they tailor dialysis prescriptions for individual patient needs.

The peritoneal equilibration test (PET) emerges as a cornerstone in the assessment of peritoneal solute transfer rates by quantifying the rate at which small solutes traverse the peritoneal membrane.<sup>50</sup> Historically, PET had been used by clinicians to tailor PD regimens according to membrane transport characteristics.<sup>50,51</sup> However, the trend of prescribing on the basis of membrane transport characteristics has become less popular, with clinicians now prioritizing patient lifestyle needs over transport kinetics. The PET also presents several limitations that challenge its routine use in clinical settings. Factors like residual volume in the peritoneal cavity, patient health status, and sample processing inaccuracies can introduce variances in PET results, making its interpretation more complex.<sup>51,52</sup> The time and resources needed to conduct PET also create logistical hurdles in routine clinical assessments. Furthermore, the 2021 ISPD guidelines recommend a shift away from the traditional practice of providing treatments based on patient categorization of Peritoneal Solute Transport Rate (PSTR).<sup>39</sup> Despite the limitations of basing treatment decisions solely on transport characteristics, assessing PSTR early in PD initiation provides important prognostic information. A faster PSTR is linked to increased risk of hospitalization and mortality,<sup>53,54</sup> highlighting that automated peritoneal dialysis (APD) may be preferable for those with rapid transport rates due to its ability to better manage fluid balance and solute. While many centers around the globe regularly perform annual retests, the advantages of repeated testing have not been conclusively established.<sup>39</sup> Given the potential for PSTR to change over time, retesting can be considered when there are noticeable clinical changes, such as signs and symptoms of volume overload where it is unclear whether volume overload is due to diminished residual kidney function, heart failure, chronic liver disease, dietary indiscretion, or a change in peritoneal membrane function.<sup>39</sup>

## LONGITUDINAL CHANGES IN THE PERITONEAL MEMBRANE

Prolonged use of PD has been associated with gradual structural and functional changes in the peritoneal

membrane. Studies examining peritoneal membrane biopsies among patients on PD have revealed significant alterations including mesothelial cell anomalies, interstitial fibrosis, elastosis, and vascular changes in the peritoneal membrane.<sup>55</sup> Functionally, there is decreased UF capacity, often due to fibrosis-related decreased osmotic conductance to glucose coupled with an expanded effective vascular surface area. Growing evidence suggests that low-pH dialysate solutions (see “[Peritoneal Dialysis Solutions](#)” later) play a pivotal role in the deterioration of peritoneal membrane functionality. This may lead to a gradual loss of osmotic conductance to glucose, which results in an uncoupling of water and solute transport across the peritoneal membrane. A key finding in low osmotic conductance to glucose is the loss of sodium sieving and collective decreased capacity for peritoneal free water clearance.<sup>56</sup>

## PERITONEAL DIALYSIS MODALITIES

PD can be offered broadly as two types of modalities: continuous ambulatory peritoneal dialysis (CAPD) and automated (machine-driven) PD (APD). Historically, these modalities have been provided to patients 24 hours per day, 7 days per week. But with increased adoption of goal-directed therapy and prescriptive approaches such as incremental PD, practice has gradually changed with patients having reduced hours or even a few days off per week<sup>57-59</sup> (see “[Goal-Directed Peritoneal Dialysis](#)” later). APD leverages technologic advances to offer patients treatment using an automated cycler. This system usually operates to instill, dwell, and drain PD fluid typically overnight while a patient sleeps and offers increased flexibility for those with busy daytime schedules. Recent iterations of APD cyclers offer improved precision and come with the capability for remote monitoring and ability to vary dwell times for each cycle and solution type. Preliminary observational data have shown that these newer cyclers with remote monitoring capabilities may be associated with improved patient quality of life, reduced emergency department visits, and lower hospitalization rates.<sup>60</sup> However, the adoption of APD cyclers is influenced by various factors such as cost, availability of specialized training, and health care infrastructure of the region. On the other hand, CAPD is delivered through manual exchanges (usually three to four per day) and its use may vary around the globe due to patient preferences, provider experience, and also the relatively lower cost compared with automated PD cyclers.<sup>12,13,21</sup> CAPD offers additional advantages for patients who want increased flexibility in traveling without the need to transport a cycler and no connection to a machine overnight to significantly lower drain pain owing to gravity versus hydraulically driven fluid drainage.<sup>61</sup>

Treatment strategies were historically guided by peritoneal solute transport status. Patients categorized as “high” or “fast” transporters would typically have a rapid dissipation of their glucose gradient and were considered better suited for the short dwells offered by APD, whereas “slow” transporters require longer dwell times to optimize solute removal. The latter group was thought to be better suited for CAPD. However, this prescriptive approach based solely on transport kinetics is increasingly viewed



as impractical for everyday use. As mentioned earlier, modern PD prescriptions take into consideration factors such as degree of residual kidney function, use, and availability of glucose-based polymer PD solutions (see “[Peritoneal Dialysis Solutions](#)” later) while prioritizing patient-centered factors like lifestyle needs, employment commitments, and other clinical considerations.<sup>62</sup> For example, mechanical considerations including back pain, a history of hernias, or a recent history of pericatheter PD fluid leak may find APD more appropriate as a means of delivering PD in the supine position with significantly lower intraabdominal pressure (IAP) compared with CAPD in the upright position. Additionally, those with daytime work commitments may prefer APD. Night time APD followed by a long daytime dwell allows for flexibility in daily schedules, catering to the needs of working professionals.<sup>57</sup>

In terms of clinical outcomes, both CAPD and APD have seen considerable evolution. Earlier studies hinted that APD had a lower risk of peritonitis due to fewer daily connections and disconnections.<sup>63</sup> However, advancements in both CAPD and APD, such as “flush-before-fill” systems, have nullified this advantage.<sup>64,65</sup> Similarly, sodium sieving is a concern with APD, which occurs during its short overnight dwell times, particularly with hypertonic (high glucose concentration) solutions with the movement of electrolyte-free water across AQP1. This leads to a drop in dialysate sodium concentration but more importantly causes transient hyponatremia, increasing the thirst drive of patients and causing them to drink water, ultimately counteracting the effects of enhanced UF.

Despite reported differences between the modalities, there is no conclusive evidence that either method is superior in terms of survival rates, quality of life, or risk of transitioning to in-center HD.<sup>66</sup> Hence the choice between CAPD and APD is ultimately dependent on patient lifestyle preferences, financial considerations, and local health care resources and practices.

## PERITONEAL DIALYSIS SOLUTIONS

The effectiveness of PD is heavily dependent on the quality and composition of the dialysate. Traditionally, the dialysate consists of three primary components: an osmotic agent to aid ultrafiltration (UF), a buffer to correct uremic metabolic acidosis, and various electrolytes to provide optimal balance of electrolytes and acid-base status. Peritoneal dialysate can be lactate or bicarbonate buffered. It comes in polyvinyl chloride (PVC) bags and is largely standardized across manufacturers. It usually excludes potassium. Dextrose, in varying concentrations, is the most commonly used osmotic agent, with amino acids and icodextrin, an oncotic agent, also being employed as alternative agents.

### DEXTROSE-BASED SOLUTIONS

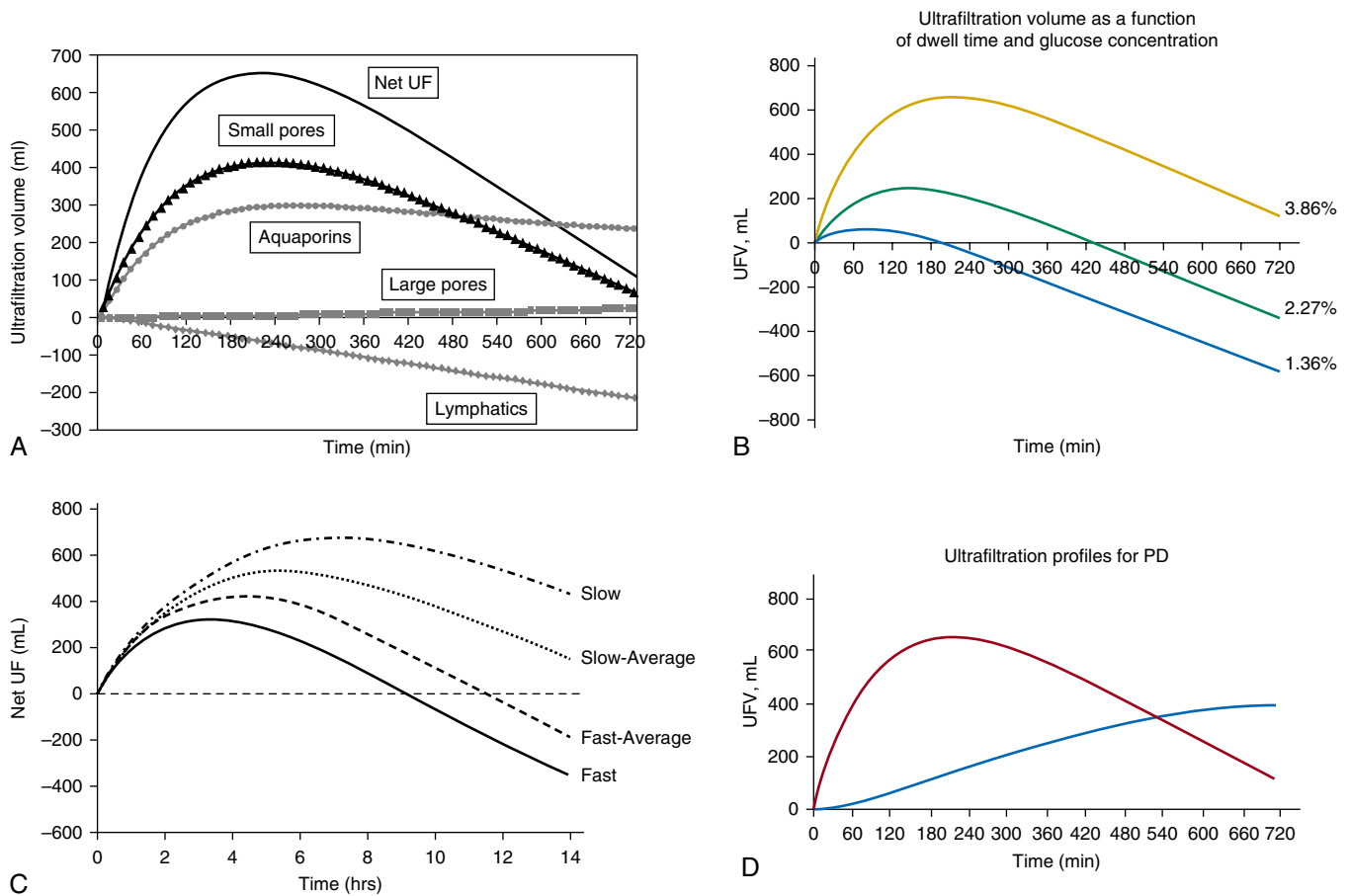
Dextrose, the primary osmotic agent in peritoneal dialysis (PD) solutions, is available in varying concentrations. In North America, solutions are labeled with the monohydrate molecular weight (MW 198 Da), offering 1.5%, 2.5%, and 4.25% options. In Europe, they are labeled by the anhydrous dextrose content (MW 180) as 1.36%, 2.27%, and 3.86%.

These represent low, intermediate, and high-concentration classes, respectively. Higher dextrose concentration solutions are used to achieve greater UF volume, particularly in patients with volume overload. Simulations by computer modeling have demonstrated that during a 12-hour dwell time, the most significant UF occurs early on, driven by a pronounced osmotic gradient ([Fig. 63.2](#)). This UF volume progressively decreases as the gradient weakens, leading to fluid being absorbed through small pores and lymphatics in the adjacent tissues. There is also a specialized 0.5% concentration solution that exists for patients who need intravascular volume expansion, but its use is confined to a handful of countries, limiting its broader application. This low-concentration solution can assist in the transfer of water and electrolytes from the dialysate back into the bloodstream, sometimes obviating the need for intravenous fluid support.

While effective, the long-term use of dextrose-based solutions presents various challenges. One significant issue is that sustained glucose absorption can lead to metabolic complications, such as hyperglycemia, hyperinsulinemia, and dyslipidemia (see “[Non-Catheter-Related Complications](#),” “[Metabolic Complications](#)”). Moreover, the osmotic gradient, which powers the UF process, tends to diminish over time. This is particularly notable during prolonged dwell times, as seen in CAPD during the day or overnight with APD.<sup>67-69</sup> Given these drawbacks and earlier mentioned challenges with glucose degradation product (GDP) generation following heat sterilization, there had been a push to develop alternative solutions. Options under consideration included PD solutions with a more stable osmotic agent suitable for extended dwell times or dextrose-based solutions engineered to minimize the formation of GDPs. The primary aim was to maintain the efficacy of treatment while mitigating its long-term drawbacks.

### LACTATE-BUFFERED SOLUTIONS

Lactate has been the traditional primary buffer used in PD solutions; however, the advancement of multichambered systems using bicarbonate as a buffer has provided a viable alternative. These systems store dextrose and buffer separately and combine them immediately before use. The multichambered design prevents the precipitation of calcium and magnesium with bicarbonate that could occur within the PVC bags with routine storage of solutions. Additionally, this innovation minimizes the generation of GDPs and enables the use of other buffers, such as bicarbonate, which is more physiologic. As a highly reactive substance, glucose readily breaks down into GDPs, which can lead to the formation of early or advanced glycation end products (AGEs). Various studies have demonstrated that exposure to GDPs may be associated with increased synthesis of vascular endothelial growth factor, mesothelial denudation, and induction of profibrotic cytokines, which in turn lead to changes in the peritoneal membrane that affect the efficiency of small solute and water transport.<sup>67-69</sup> The production of GDPs is influenced by factors such as pH, temperature, and time, particularly when PD solutions undergo heat sterilization.<sup>70</sup> As a result of the aforementioned hazards of GDPs, there has been growing interest for more “biocompatible” solutions,<sup>71</sup> described briefly as follows.



**Fig. 63.2** (A) Computer model depicting ultrafiltration (UF) across the peritoneal membrane using a 3.86% glucose solution demonstrates that, following the dissipation of the osmotic gradient at 240 minutes, fluid reabsorption occurs via small pores and lymphatic pathways. Due to the correlation of solute transport with the area of small pores, membranes with higher transport capabilities absorb more fluid. (B) The ultrafiltration volume (UFV) over time is represented for different glucose concentrations: 3.86% (yellow), 2.27% (green), and 1.36% (blue), using a three-pore model simulation. (C) The graph shows a comparison between the ultrafiltration profile for 7.5% icodextrin (blue line) and a 3.86% glucose solution (red line), both simulated with the three-pore model. (D) Variations in UF are observed for different peritoneal membrane solute transfer rates using a 2 L 2.5% dextrose solution. Patients with rapid solute transfer rates in their peritoneal membranes experience reduced maximum UF and a quicker onset of fluid reabsorption. (Adapted with permission from [A],<sup>208</sup> [C],<sup>209</sup> and [B and D].<sup>210</sup>)

### LOW-GLUCOSE DEGRADATION PRODUCT SOLUTIONS

The multichambered, or low GDP neutral pH, solutions had been proposed as an alternative to overcome some of the limitations of traditional lactate-buffered solutions. They offer a more neutral pH, potentially reducing the discomfort associated with infusion. They also enable the use of heat sterilization methods that are less likely to produce harmful GDPs. However, the clinical benefits of these newer solutions have been a subject of debate.

The efficacy and safety of biocompatible solutions remain topics of contention. The balANZ study, a randomized controlled trial (RCT), reported that the use of low GDP neutral pH solutions resulted in a decreased incidence of peritonitis compared with those using conventional solutions.<sup>72</sup> However, subsequent RCTs have produced varied findings, leading to uncertainties about the true clinical advantages of these biocompatible solutions.<sup>73-75</sup> Despite a Cochrane review of 15 studies involving 835 patients showing significant preservation of residual kidney function with the use of neutral pH, low-GDP PD solutions, its widespread adoption is still limited,

underscoring the ongoing controversy in this area.<sup>74</sup> Overall, systematic reviews have offered inconclusive evidence about the overarching benefit of low GDP neutral pH solutions in PD.<sup>71,74</sup> As a result, these solutions are now often reserved for patients experiencing infusion pain with conventional solutions. These solutions may also lead to better preservation of peritoneal membrane function compared with lactate-buffered solutions.<sup>76</sup> While low GDP solutions may offer many benefits, studies in pediatric patients have shown they can still induce early peritoneal inflammation, fibroblast activation, and angiogenesis, which affect membrane transport function.<sup>77</sup>

Electrolytes, specifically sodium, calcium, magnesium, and chloride, are fundamental to PD solutions. Standard PD solutions have slightly lower sodium concentrations than those used for HD, which facilitates diffusive sodium removal. Although preliminary findings suggest that reducing sodium concentration in dialysate could enhance sodium removal by increasing the sodium gradient between blood and dialysate, these promising results have yet to

translate into changes in clinical practice, as no definitive benefits from using low-sodium PD solutions have been documented to date.<sup>78</sup>

### DEXTROSE-FREE SOLUTIONS

In recognizing the shortcomings of dextrose-based solutions, attention had shifted toward other osmotic agents, namely amino acids and icodextrin. Icodextrin, an iso-osmolar glucose polymer with an osmolality of approximately 280 mOsm/L, is offered as a 7.5% solution.<sup>79</sup> It shares an electrolyte profile with dextrose-based solutions. Its molecular weight, which ranges between 13 and 19 kDa, coupled with its oncotic effect, significantly amplifies the osmotic efficiency of the solution, outperforming dextrose-based alternatives.<sup>79</sup> This characteristic leads to prolonged UF over 12 hours, making icodextrin especially suited for extended dwell exchanges.<sup>80</sup> For example, it is preferred for daytime dwells in patients on APD and nighttime dwells for CAPD, as its large size prevents crossing the peritoneal membrane, thus maintaining a prolonged oncotic effect across the membrane.

Currently, icodextrin is approved for once-daily administration. However, some observational studies hint that administering it twice daily can be beneficial for certain patients. Additionally, meta-analytic data have demonstrated that icodextrin may improve peritoneal UF and reduce risk of volume overload in patients.<sup>74</sup> An essential point for medical professionals to note is the potential for maltose buildup in the extracellular compartment when using icodextrin. This accumulation can trigger a shift of water from the intracellular space, resulting in mild hyponatremia.<sup>81</sup> Although the body can gradually absorb and metabolize icodextrin into maltose, there is no documented maltose-related toxicity. Nonetheless, maltose can interfere with specific assays used for blood glucose measurements, yielding inaccurately high glucose readings.<sup>82</sup> Therefore when treating patients with icodextrin, it is important to consider alternative blood glucose monitoring methods or ensure that the chosen glucometer is compatible with icodextrin and does not misinterpret maltose as glucose.<sup>82</sup>

Amino acid–based PD solutions were developed to counteract protein-energy malnutrition, a prevalent challenge for patients on dialysis.<sup>83,84</sup> Amino acid–based solutions are primarily used for patients suffering from inability to maintain consistent enteral protein intake or who suffer from severe malnutrition or in situations demanding strict metabolic control. They serve as a nutritional supplement in these select cases while also offering the added advantage of minimizing the overall exposure of dextrose to the peritoneal membrane. These solutions contain a 1.1% amino acid blend, mirroring the osmolality seen in a 1.36% dextrose solution. While amino acid dialysate solutions can be used to enhance nutrition, particularly in high-risk patients, observational data have not found a significant impact on patient survival. As a result, their effectiveness in real-world scenarios has unfortunately fallen short of expectations.<sup>85</sup> Due to potential risks including acidosis and elevated plasma urea levels from absorbed amino acids, their use is restricted to a single exchange per day.<sup>86</sup> Presently, these solutions are available in some centers around the globe and are primarily used for glucose-sparing purposes.

Studies exploring dextrose-free PD solutions, particularly those using L-carnitine and xylitol, have shown encouraging outcomes and clinical trials are under way.<sup>87</sup> Initial observations indicate a favorable safety profile, alongside efficient solute removal and effective peritoneal ultrafiltration.<sup>88</sup> These preliminary results suggest potential benefits in the management of PD patients and underscore the need for more extensive, controlled trials to confirm their clinical efficacy and safety.

## PERITONEAL DIALYSIS ACCESS

### HISTORY AND EVOLUTION OF PERITONEAL DIALYSIS CATHETER TYPES

The success of PD is heavily dependent on creating reliable, safe, and functional permanent access to the peritoneal cavity. PD catheters are often considered the “lifeline” for patients receiving this type of treatment. Disruption to this “lifeline” can lead to a premature end to PD therapy and transition to HD. This is supported by observational data demonstrating that about 20% of PD discontinuations are attributable to catheter-related complications.<sup>89</sup> Given the importance of safe and reliable peritoneal access, PD catheters have undergone significant changes over the past several decades aiming to improve patient outcomes, reduce complications, and enhance overall comfort. PD access of the 1920s was characterized by simpler catheters, often made of rigid glass or metal tubes that were directly inserted into the peritoneal cavity. These catheters did not cater to longevity on PD as they were commonly plagued with complications such as infections, leaks, bowel perforation, and other damage to internal organs. In the contemporary period, PD catheters have undergone significant advancements.

Modern catheters consist of three main parts: the intraperitoneal, tunneled, and extraperitoneal segments. The intraperitoneal segment of the catheter is within the abdominal cavity, with the most distal end in the rectovesical pouch. The PD catheter traverses the abdominal wall into a subcutaneous tunnel with the most proximal end resulting in external protrusion at an exit site.<sup>90</sup> The extraperitoneal segment is commonly augmented with two Dacron (polyethylene terephthalate) cuffs, delineated as the deep (preperitoneal) and superficial (subcutaneous) cuffs, which are both integral to overall catheter functionality. Some catheter designs have a deep single cuff only. The cuffs facilitate anchorage via promoting tissue ingrowth and also promote a barrier mechanism that curtails the risk of infiltration of bacteria. The optimal positioning of the deep cuff is within the rectus muscle sheath of the anterior abdominal wall, facilitating tissue encapsulation and thereby mitigating migration risk.<sup>90</sup> In contrast, the superficial cuff, ideally situated 2 to 4 cm proximal to the exit site, serves as a “piston,” precluding microbial ingress into the subcutaneous trajectory.

Among the pioneering designs in this field is the Tenckhoff catheter, conceptualized by Dr. Henry Tenckhoff in 1968.<sup>91</sup> This catheter heralded a significant paradigm shift in PD catheter designs with the introduction of a Silastic tube.<sup>91</sup> Silicone, the primary constituent material of the catheter, demonstrates increased durability, flexibility,

and biocompatibility, properties that mitigate the incidence of adverse tissue reactions, a notable challenge with earlier versions of PD catheters (see Fig. 63.3 for different catheter designs).

Beyond the original Tenckhoff design, alternative configurations like the swan-neck catheter have since been conceptualized. This variant is typified by a pronounced 180-degree angulation interposed between the cuffs, a morphologic alteration engineered to enhance positional stability and curtail risks associated with internal tip migration. Further classifications based on tip permutations within the swan-neck design include the coiled, straight, and bifurcated silicone disk variants, namely the Toronto-Western or Oreopoulos-Zellerman.<sup>91</sup> These particular designs have been designed to counter the physiologic phenomenon of wrapping omentum around the catheter, thereby maintaining intraperitoneal stability of the catheter within the pelvis.<sup>91</sup> These iterations forgo the internal cuff in favor of a composite of felt disk and silicone bead.

A modern Tenckhoff catheter now includes a radiopaque stripe, allowing for straightforward visualization on plain film radiographs, which aids in determining its position within the intraperitoneal cavity. It is also punctuated with multiple 0.5-mm lateral apertures, spanning a length ranging between 10 and 15 cm on its intraperitoneal segment.<sup>91</sup> Today, Tenckhoff catheters are primarily produced in two designs, the double-cuffed straight-tip and double-cuffed coiled tip. It has been postulated that the

coiled-tip design results in diminished inflow discomfort due to a superior and gentler dispersion of dialysate fluid, negating the concentrated jet flow seen with its straight-tip designs.<sup>91</sup> However, empirical data sourced from multiple studies analyzing outcomes and complication rates between the two catheter-tip variants have yielded mixed results, and thus a consensus on the categorical superiority of one design over another is lacking.<sup>92-94</sup> As a result, either catheter tip is appropriate for use among patients on PD and use should depend on local availability.

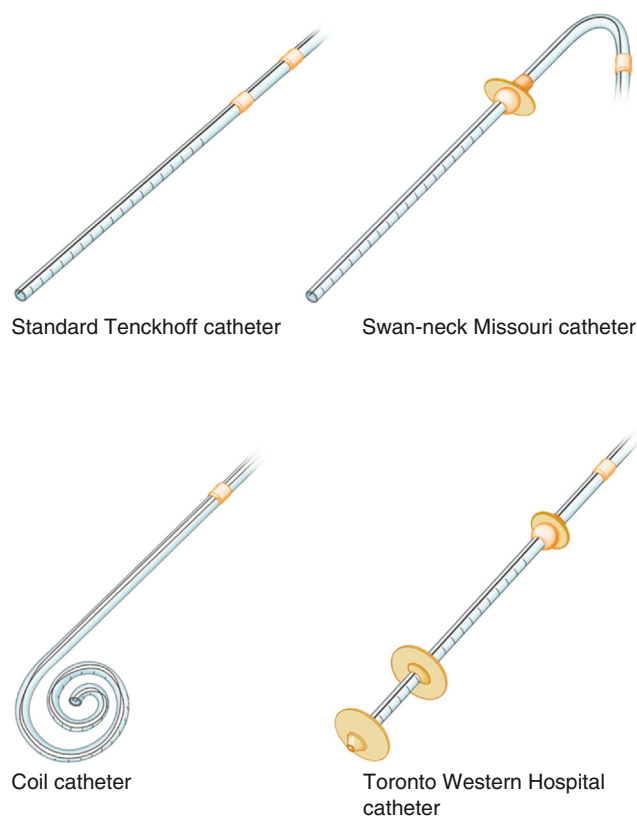
Additional variations in catheter designs exist, and these include extended-length catheters. Extended-length catheters are designed for specific patient conditions including obesity, incontinence, intestinal stomas, gastrostomy tubes, or suprapubic urinary catheters.<sup>90,95</sup> These catheters allow for the positioning of exit sites in areas that are visible and easily accessible to patients or care providers in the upper abdominal area or presternal area, allowing for appropriate care and ultimately reducing the risk of exit site infections.

## PREPARING FOR PERITONEAL DIALYSIS CATHETER INSERTION

For optimal PD catheter placement, preoperative preparations are essential to ensure successful insertion and mitigate complications. A thorough assessment is crucial for identifying previous surgical scars and hernias, as well as determining the ideal location of the PD catheter exit site.<sup>90</sup> Minimizing infection risk entails selecting the most suitable exit site, factoring in patient body habitus, abdominal creases, stoma locations, and other factors that can potentially obscure the exit site, preclude proper routine exit site care, or increase risk of infection.<sup>90,96</sup> It is important to conduct the abdominal assessments with the patient in a seated position to rule out prominent skin folds that may be less apparent compared with when a patient is in a supine position. This detailed assessment guides the selection of catheter length and the potential need for an extended exit site. Excessive skin folds may warrant an upper abdominal or a presternal extended catheter exit site.<sup>90</sup>

In terms of catheter insertions, multiple techniques exist, from percutaneous needle-guidewire methods (with or without adjunct fluoroscopic and ultrasound imaging) to open surgical dissection and surgical laparoscopy.<sup>90</sup> The chosen technique depends largely on available resources and local expertise, with a global trend favoring laparoscopic and percutaneous methods.<sup>97</sup> Despite differences in technique between these approaches, there are important similarities that align with best practice recommendations as outlined by the ISPD. These recommendations propose that catheters be inserted using a paramedian approach through the rectus muscle and a purse-string suture placed around the deep cuff.<sup>90</sup> In placing a purse-string suture around the deep cuff, the risk of leak can be reduced by ensuring that the cuff is well embedded into the muscle.<sup>98,99</sup> Additional technical recommendations include inserting the catheter with posterior rectus sheath tunneling, where possible, to reduce the risk of catheter-tip migration. Lastly, it is recommended that the exit site be, at minimum, 2 cm away from the superficial cuff.<sup>90</sup>

The 2019 ISPD recommendations for maintaining optimal peritoneal access and the most recent 2022 ISPD



**Fig. 63.3** Differences in the intraperitoneal and extraperitoneal segments between currently available peritoneal dialysis catheters used in clinical practice.



peritonitis guidelines recommend a preoperative dose of systemic antibiotics.<sup>90,100</sup> A preoperative dose of a first- or second-generation cephalosporin or vancomycin is typically administered for patients undergoing a PD catheter insertion.<sup>100</sup> The recommendations proposing this approach are supported by systematic review data that reveal the efficacy of prophylactic systemic antibiotics before PD catheter insertion.<sup>101</sup> Although first-generation cephalosporins may be marginally less effective than vancomycin, they are still frequently used due to concerns about the development of vancomycin resistance.<sup>100</sup> Ultimately, antibiotic choice is influenced by local antimicrobial resistance patterns and patient-specific allergy profiles.

## ADVANCED LAPAROSCOPIC INSERTION

The surgical laparoscopic approach has now become the favored method among clinicians and is often considered the gold standard for surgical PD catheter insertions. This preference stems from the numerous advantages offered by providing direct visualization of the intraperitoneal cavity that allows operators to introduce specific interventions, thereby heightening the overall success rate.<sup>90,96</sup> An important advantage with advanced laparoscopic insertion is the ability to manage redundant omentum in the intraperitoneal cavity. If the omentum is left unchecked and in a state of redundancy, it can significantly impede the optimal functioning of the catheter. The laparoscopic approach allows an operator to suture redundant omentum upon itself or to an adjacent organ in a procedure called an omentopexy.<sup>102,103</sup> This process not only resolves the immediate redundancy concern but ensures a smoother operational pathway for the catheter. The laparoscopic surgical approach may also be well suited for patients with small hernias and provides the opportunity for skilled operators to identify defects that can be fixed at the same time of catheter insertion or thereafter. Data from the North American PD Catheter Registry highlights the importance of adhesiolysis as an adjunct procedure.<sup>104</sup> Patients with a history of adhesions face an increased risk of catheter flow dysfunction and abdominal pain. Therefore lysis of adhesions during catheter placement can be crucial in improving outcomes in this population. Ultimately, a history of abdominal surgery should not automatically disqualify patients from PD catheter insertion, as observational data support the conclusion that prior abdominal surgery is not a contraindication for PD.<sup>105,106</sup>

## PERCUTANEOUS INSERTION

While the advantages of advanced laparoscopic insertion are numerous, it is not always feasible or indicated for every patient or clinical setting. The percutaneous insertion method provides a compelling alternative, particularly in environments with limited resources or where a general anesthetic may not be tolerated. This approach applies the principles of a modified Seldinger technique by using a combination of a needle, guidewire, and in some instances an ultrasound.<sup>90,107</sup> It is particularly well suited for patients with no prior abdominal surgical history and no abdominal hernias. Fluoroscopy can be

used as an adjunct with ultrasound to minimize risk of injury and confirm catheter placement in the rectovesical pouch.<sup>107</sup>

An additional advantage of a percutaneous approach is that it creates one point of entry through the peritoneal cavity and allows for the PD catheter to be used immediately, whereas a surgical laparoscopic insertion results in additional defects in the peritoneal cavity with the need for laparoscopic ports to allow for the use of specialized equipment.<sup>90</sup> The need for laparoscopic ports creates additional defects in the peritoneal cavity as potential sources of fluid leaks. The presence of these additional holes in the peritoneal cavity requires time for those areas to heal.

Emerging data from the United Kingdom highlight significant benefits of percutaneous PD catheter insertions. The UK Cath study was a prospective, multicenter cohort investigation conducted in 44 out of 72 UK dialysis centers, focusing on patients undergoing their first catheter insertion, with 20 centers also participating in the PDOPPS.<sup>25,108</sup> This study reinforced prior evidence that a physician-led medical pathway enhances access to PD and suggested that outcomes are similar or improved under this approach. This improvement is likely due to the use of medical catheter insertion techniques in centers with a stronger focus on PD and more experience with this modality.

In the wake of the COVID-19 pandemic, the medical community observed a notable uptake in the adoption of percutaneous PD catheter insertions.<sup>109</sup> Several reasons can be attributed to this change in practice. First, the pandemic exerted unprecedented pressure on health care infrastructure, leading to a significant reduction in available resources, including operating rooms and specialized surgical equipment.<sup>110</sup> Secondly, there was a need to reduce prolonged hospital stays and patient exposure risk in hospital environments, particularly in congregate settings where the risk of COVID-19 transmission was higher.<sup>110,111</sup> The percutaneous approach, which can be performed at the bedside with minimal resources and allowed patients to be on home dialysis, ultimately emerged as a viable alternative in reducing risk of COVID-19 infection for patients.

## COMPLICATIONS OF PERITONEAL DIALYSIS

### NON-INFECTIOUS COMPLICATIONS

#### CATHETER-RELATED

Catheter-related complications in PD can have a significant impact on patient quality of life and overall effectiveness of therapy. The early identification and prompt management of these complications may reduce the likelihood of a premature transition to HD. Many non-infectious catheter-related complications are predominantly related to flow dysfunction. Flow dysfunction in PD is multifactorial and can be further categorized into one-way or two-way obstruction.<sup>90</sup> Two-way obstruction can be attributed to the suboptimal evacuation of the dialysate fluid, due to visceral structures within the pelvic anatomy interacting with the catheter tip and covering the apertures along the distal intraperitoneal segment. This obstruction can occur due to extrinsic compression (due to omentum) or

kinking of the catheter itself, intraluminal obstruction, and catheter entrapment.<sup>90,112</sup>

### CATHETER MIGRATION AND ENTRAPMENT

PD catheter entrapment, a notable and concerning complication, arises predominantly when the catheter is ensnared within the omental matrix or tethered by intraperitoneal adhesions.<sup>103</sup> It can also occur with other structures in the pelvis, and there have been documented reports of fallopian tube entrapment leading to flow dysfunction in the literature.<sup>113,114</sup> This phenomenon leads to occlusion of the holes along the side of the catheter, manifesting as either unidirectional or bidirectional flow dysfunction. Regardless of the type of entrapment, it usually requires advanced laparoscopic intervention to allow for a release of the entrapment or exchange for a new PD catheter.<sup>115</sup> Patients with suspected entrapments should not have the catheter exchanged blindly or use a percutaneous approach as there is significant risk for visceral and organ injury without direct visualization.

On the other hand, extrinsic compression is a type of dysfunction that often emerges because of pathophysiologic conditions such as bladder distension or constipation.<sup>90</sup> Constipation is often identified as a leading cause of one-way outflow obstruction due to the pressure exerted by the distended colon, impeding the patency of the catheter by obstructing the apertures on the distal end. The diagnostic workup for this condition typically involves an abdominal radiograph to assess significant fecal loading and catheter-tip position. Management in such cases often depends on the underlying cause but usually requires treatment with a laxative regimen to relieve the obstructive effect on the catheter. It should be important to recognize that excessive external compression in some cases can result in two-way obstruction in PD catheters.

### TWO-WAY OBSTRUCTION FIBRIN, CLOTS, AND CATHETER KINK

On another note, a significant subset of PD catheter flow dysfunction emanates from intraluminal obstructions. These obstructions, which manifest as impediments to both the instillation and drainage phases of PD, can arise from mechanical kinks or intraluminal debris such as fibrin.<sup>115</sup> The development of fibrin is commonly seen after an inflammatory process such as peritonitis or even after a catheter is newly inserted. The therapeutic approach for this complication is a stepwise approach of irrigating the catheter with saline flushes and a combination of intraluminal heparin or tissue plasminogen activators when fibrin is suspected.<sup>116</sup> In cases where flow remains obstructed despite the aforementioned interventions, imaging may be warranted to rule out any catheter kinks. In cases where catheter kinks are absent, fluoroscopic-guided manipulation may be required to restore catheter luminal patency.<sup>115,116</sup> Additionally, a small single-center Canadian observational study suggested that fluoroscopic manipulation of catheters may also be considered in cases where conservative approaches to restore function have been ineffective.<sup>117</sup> However, the success rate of fluoroscopic manipulation for catheters that have

migrated to the upper abdomen or have primary failure is relatively low, due to significant anatomic issues like omental wrapping and compartmentalization by adhesions.<sup>117</sup>

## NON-CATHETER-RELATED COMPLICATIONS

### METABOLIC COMPLICATIONS

#### HYPERGLYCEMIA

Metabolic complications in PD are primarily associated with systemic glucose absorption. This absorption may contribute 400 to 1200 kcal/day and can lead to hyperglycemia.<sup>118</sup> Managing hyperglycemia can be challenging for patients on dialysis. Although biguanides are contraindicated for patients with advanced kidney disease, other antiglycemic agents including certain sulfonylureas and glucagon-like peptide 1 receptor agonists can be used in clinical practice. Despite the theoretical benefits of SGLT2 inhibitors in preserving residual kidney function and reducing peritoneal fibrosis,<sup>119</sup> there are limited data to date on safety and efficacy within the PD population, and SGLT2 inhibitors lose their hypoglycemic efficacy as GFR drops below approximately 45 mL/min/1.73 m<sup>2</sup>. Intraperitoneal insulin offers convenience for patients but may increase the risk of bacterial contamination with injection of dialysate bags and focal hepatic steatosis.<sup>120</sup> As a result, the subcutaneous administration of insulin is the preferred method for patients on PD who require insulin.

The IMPENDIA (Improved Metabolic control of Physioneal, Extraneal, Nutrineal [P-E-N] vs. dianeal only in DIabetic CAPD and APD patients) randomized trial underscored that while glucose-sparing treatments in patients on PD with diabetes enhance metabolic outcomes—manifested by lowered HbA1c, VLDL cholesterol, triglycerides, and apolipoprotein B—there is a notable rise in adverse events, likely linked to volume overload.<sup>121</sup> This includes a marked decrease in serum albumin and a higher incidence of serious adverse events and mortality in the glucose-sparing group, underscoring the need for vigilant fluid volume status monitoring with such regimens.

#### DYSLIPIDEMIA

Patients on PD often show higher levels of total and LDL cholesterol, apolipoprotein B, lipoprotein (a), and triglycerides and lower levels of high-density lipoprotein cholesterol compared with those on HD, possibly due to systemic glucose absorption and peritoneal protein losses.<sup>122</sup> While statins can safely lower serum cholesterol in patients on dialysis, their effectiveness in reducing cardiovascular mortality is not yet proven.<sup>123</sup> The Study of Heart and Renal Protection (SHARP), which included patients receiving PD, found a reduced risk of cardiovascular events with simvastatin-ezetimibe treatment but no significant effect on cardiovascular mortality and an attenuated benefit in patients on dialysis relative to those with non-dialysis-requiring CKD.<sup>124</sup> Further research is needed to understand the efficacy of statins in reducing the risks of cardiovascular events and mortality in patients

receiving dialysis, especially those on PD where data are particularly scarce.

## ELECTROLYTE DISTURBANCES

Patients frequently exhibit disorders of mineral metabolism while on PD. Hypercalcemia can arise due to treatment with calcitriol or active vitamin D analogs, calcium-rich dialysate solutions, or calcium-based phosphate binders.<sup>125,126</sup> Hypercalcemia may reflect the underlying cause of end-stage kidney disease, such as multiple myeloma or less commonly sarcoidosis, or more commonly can reflect low turnover bone disease. Strategies to mitigate the risk of hypercalcemia include the use of lower calcium concentrations in dialysate along with noncalcium phosphate binders. Hyperphosphatemia can be associated with mortality, cardiovascular events, and fractures in patients receiving dialysis including those on PD.<sup>127</sup> Typically, hyperphosphatemia is less severe in patients on PD relative to those on hemodialysis, in part owing to better preservation of residual kidney function. Non-calcium-based phosphate binders may be particularly well suited for patients on PD, as calcium-based binders tend to cause constipation and patients on PD. Ferric citrate is among the more potent intestinal phosphate binders and can serve as an iron supplement, potentially abrogating the need for intravenous iron supplementation to support erythropoiesis.<sup>128</sup> Control of hyperphosphatemia may be improved by the addition of the phosphate absorption inhibitor tenapanor, a sodium-hydrogen exchanger 3 inhibitor, which reduces passive intestinal phosphate absorption, and can result in softening of stools.<sup>129</sup>

Patients on PD are at an increased risk of developing hypokalemia due to the nature of potassium-free dialysate solutions, which enhances the diffusive gradient for its clearance. Additionally, patients will also achieve convective clearance, particularly in the presence of large UF volumes, a phenomenon referred to as solvent drag. Hypokalemia frequently occurs in patients with inadequate nutrition and is exacerbated by high doses of loop diuretics (with or without thiazides) prescribed to patients with residual kidney function. In terms of management, patients can typically normalize serum potassium concentrations by liberalizing diets typically recommended for patients with kidney failure, particularly by increasing intake of potassium-rich fruits and vegetables, which not only constitute excellent sources of dietary potassium but can provide additional dietary fiber, often aiding in the prevention of constipation. For patients with hypokalemia unwilling or unable to materially increase dietary potassium intake, oral potassium supplements can be added to the medication regimen. In a randomized trial from Thailand with 167 participants, it was found that, compared with reactive potassium supplementation at serum levels below 3.5 mEq/L, a protocol-driven oral potassium regimen aiming to keep serum potassium between 4 and 5 mEq/L could lower the risk of peritonitis in PD patients with hypokalemia.<sup>130</sup> This underscores the importance of managing hypokalemia in this population. Patients with hypokalemia on PD should not be managed with the addition of potassium to the dialysate solutions due to increased risk of contamination and peritonitis. Hyperkalemia is rarer, often due to factors like renin-angiotensin system blockade or dietary excess, and is typically mild if dialysis is consistent.

Hyponatremia is another common electrolyte abnormality seen in PD, often the result of excessive water intake. Hyponatremia may also be associated with malnutrition (hypoosmolar).<sup>131,132</sup> Certain cases of hyponatremia (hyperosmolar) may arise secondary to hyperglycemia or icodextrin administration, inducing a hyperosmolar state. Additionally, in the context of severe malnutrition, a hypoosmolar variant of hyponatremia can manifest.<sup>131,132</sup> Hypernatremia is rare but can occur in older patients with reduced thirst sensation or limited water access. Intense and frequent dialysis using hypertonic dialysate might cause hypernatremia through sodium sieving, a concept discussed in earlier sections. Hypernatremia due to sodium sieving can be prevented by incorporating longer dwell periods in the 24-hour dialysis cycle to facilitate sodium-coupled fluid removal.

Hypomagnesemia has been linked with poorer nutritional status, suggesting that malnutrition may be a significant contributor. Additionally, observational data have suggested that frequent use of hypertonic dialysate solutions may also lead to hypomagnesemia in patients on PD.<sup>133</sup>

## FLUID LEAKS

### PERITONEAL DIALYSIS HYDROTHORAX

Hydrothorax complicates PD with an estimated incidence between 1.6% and 10%.<sup>134,135</sup> Hydrothorax occurs when dialysate fluid from the intraperitoneal cavity enters the pleural space as a result of pleuroperitoneal communication, with the majority of cases occurring on the right side.<sup>136</sup> The prevailing theory is that most pleuroperitoneal leaks occur on the right side due to a congenitally porous diaphragm, with the liver functioning as a piston and the ascending colon exerting upward peristaltic motion, driving fluid through these openings. Patients may present with dyspnea and, in more severe instances, respiratory distress with associated lung collapse. However, approximately 25% of patients present with no symptoms and hydrothorax is identified incidentally on imaging.<sup>134</sup> Although the underlying mechanism by which the dialysate fluid enters the pleural space is thought to be related to pleuroperitoneal communication as mentioned earlier, increases in IAP can also play a significant role.

In terms of diagnostic evaluation, chest radiograph commonly identifies hydrothorax and should be performed with patients in a lateral decubitus position, which is useful in ensuring that the associated pleural effusion is not loculated. Patients presenting with symptomatic PD hydrothorax may benefit from therapeutic thoracentesis. Diagnostic thoracentesis allows for fluid analysis, which may reveal transudative effusions associated with a high glucose concentration and extremely low protein levels. However, in cases where a non-dextrose-based solution is used, such as icodextrin, glucose concentrations in the effusion will be similar to that of the plasma. Additionally, glucose concentrations in a chronic pleural effusion may be low due to pleural glucose absorption. As a result, the absence of elevated glucose in a pleural effusion does not rule out pleuroperitoneal leak. Another diagnostic tool that had been historically employed was the instillation of methylene blue into the peritoneal cavity, which would subsequently appear in the pleural fluid. However, this approach has

fallen out of favor due to an increased risk of chemical peritonitis with methylene blue use in the peritoneal cavity. Another noninvasive approach in centers equipped with the appropriate resources is the use of peritoneal scintigraphy. This uses technetium-99m labeled albumin added to dialysate solution, and the patient is evaluated using a gamma camera to determine if any of the labeled albumin is identified in the pleural cavity.<sup>137</sup> A computed tomography (CT) peritoneogram (see hernia section for details) can also be used but may lead to inconclusive results, especially if the pleuroperitoneal defect is small, not allowing for dispersion of iodinated contrast to be seen in the pleural space. The CT peritoneogram uses the presence of iodinated contrast in the dialysate. To increase the diagnostic sensitivity of this study, patients should have fluid dwell for at least 2 to 3 hours or as long as possible to allow for dispersion of fluid through any defects that may be present.<sup>138</sup>

Though there are some reports of temporary cessation of PD resulting in healing of the hydrothorax, definitive management of a PD hydrothorax is often required.<sup>139</sup> This entails closure of the existing pleuroperitoneal communication using video-assisted thoracoscopic surgery (VATS) to allow for surgical correction of diaphragmatic defects coupled with the use of a sclerosing agent for targeted pleurodesis. VATS allows direct visualization of possible diaphragmatic defects and can lead to greater success in preventing recurrence of PD hydrothorax.<sup>136</sup> However, patients will typically require respite of several weeks from PD to allow for effective healing of closures. In cases where surgical closure of pleuroperitoneal communication is not feasible, an attempt at bedside pleurodesis may be warranted; otherwise, the patient is likely to require transition to HD.

#### PERICATHETER LEAKS AND GENITAL EDEMA

Practitioners should also be familiar with another important complication in the form of pericatheter leaks. These leaks usually occur as an early complication after catheter insertion or manipulation but could also happen at any point while on PD. Additionally, ports used during advanced laparoscopic insertion may also be sources of leaks that should be considered. An early sign of a pericatheter leak is the presence of reduced UF volumes, followed by progressive weight gain with swelling in the abdominal subcutaneous tissues or genital areas.<sup>140</sup> Many of these leaks can be managed supportively by adjusting PD therapy with reduced fill volumes in a supine position. However, if leaks are persistent despite these approaches, then additional imaging in the form of CT peritoneogram may be considered to determine if any sources of the leak require surgical correction.

Genital edema in patients on PD is a significant complication. It typically arises when dialysate travels through a patent processus vaginalis, leading to hydrocele and potential scrotal or labial wall edema, or via an abdominal wall defect near the catheter tract, causing edema in the genital area. To identify the path of the leak, a CT peritoneogram is recommended to distinguish between fluid tracking through a patent processus vaginalis (indirect hernia) or an abdominal wall defect (direct hernia). Treatment requires stopping PD, with bed rest and scrotal elevation as supportive measures. APD performed in a supine position with low fill volumes and no daytime dwell

can effectively lower IAP and prevent the recurrence of leaks. If leaks persist, transitioning to HD may be necessary, either as a temporary or permanent measure, based on the feasibility of surgically repairing the leak sources. Additionally, if there is a patent processus vaginalis, then definitive surgical repair is required to close the defect, as explained in the next section.

#### HERNIAS

Patients receiving PD are at an increased risk of developing hernias due to elevations in IAP. IAP is influenced by age, body size and composition, sitting position, dialysate volume, and prior abdominal surgery.<sup>141-143</sup> Other risk factors for hernia development include female patients and those with a diagnosis of polycystic kidney disease.<sup>143</sup> The IAP is highest when a patient sits, decreases when a patient stands, and is lowest when he or she is supine. Observational data indicate that umbilical and inguinal hernias are the most common types encountered in patients on PD.<sup>144</sup> Many of these hernias, particularly in the inguinal or periumbilical areas, predate initiation of PD but become more pronounced due to increased IAP from the volume of intraperitoneal dialysate. Additional data have noted that about 10% to 25% of patients on PD will eventually develop hernia(s) during their time on therapy.<sup>135,144</sup> New hernias can also emerge in different areas including the site of catheter insertion.

Treatment of hernias varies and depends on the type of hernia. While most hernias require surgical intervention, a conservative approach might be reasonable for certain patients, especially older patients including those too frail for surgery. Small hernias have a high risk of bowel incarceration and should be repaired surgically. If a hernia is surgically repaired, it is imperative to have PD prescriptive strategies to reduce intraabdominal pressure, which may decrease the likelihood of recurrence. Patients may require a period of about 3 to 4 weeks to allow for effective healing. If a patient does not have significant residual kidney function, he or she may require a brief transition to HD to allow healing after surgery. In patients with ample residual kidney function, an alternative approach to consider would be temporary (i.e., 48-hour holding of PD in the immediate postoperative period) followed by low-volume exchange supine PD.<sup>145</sup> The use of mesh in hernia repairs is generally preferred to minimize recurrence risk, but the decision to employ mesh depends on the hernia type, location, along with the judgment and expertise of the surgeon. Location of the mesh exposed to the peritoneal surface may preclude the use of bridging HD with the need for a 4- to 6-week period of peritoneal healing at the mesh site, though in most cases repair can be achieved with deeper mesh placement.

#### ENCAPSULATING PERITONEAL SCLEROSIS

Encapsulating peritoneal sclerosis (EPS) is a rare yet severe complication seen in patients who have been on PD for a prolonged period.<sup>146</sup> Historically, the incidence rate varied from 1.4 to 13.6 cases per 1000 patient-years around the globe but now is much less common, as highlighted in the 2017 ISPD position paper.<sup>147-149</sup> In this position paper, the ISPD emphasized that most patients undergoing long-term PD do not experience EPS with current data reporting an incidence between 0.7 and 9.5 episodes per 1000 patient-years.<sup>149</sup> In EPS, the peritoneal membrane undergoes



significant sclerosis, causing the intestines to become encased or “cocooned,” leading to major disruptions in intestinal function including major motility disorders. As a result, patients may develop a myriad of gastrointestinal symptoms and complications including nutrient malabsorption, intestinal obstruction, fluid accumulation in the abdomen, appetite loss, and progressive weight decline. Patients often exhibit systemic inflammation marked by mild fever, low serum albumin concentrations, and increased inflammatory cytokines including tumor necrosis factor- $\alpha$  and interleukin 6.<sup>147,148,150,151</sup> The diagnosis of EPS requires signs of intestinal obstruction combined with evidence of bowel encapsulation, which can be determined through methods like CT scans or pathologic examinations.<sup>147</sup> CT scans are particularly reliable for making the diagnosis in a variety of clinical scenarios with available CT scoring systems for EPS diagnosis.<sup>152</sup> However, if a patient undergoes surgery, pathologic confirmation becomes possible but extra caution is essential during surgical procedures to avoid the risk of bowel perforation.<sup>147</sup>

Although the exact cause of EPS remains unknown, several contributing factors have been identified and led to the proposal of a two-hit hypothesis model by many experts.<sup>147,153</sup> The first hit is related to exposure of PD components, such as the catheter; hyperosmotic dialysate solutions, which result in mesothelial cell damage; and subsequent peritoneal sclerosis. The second hit is suspected to be related to PD-independent factors such as genetic predispositions and exposure to certain medications. Despite the emergence of a two-hit hypothesis, the broad risk factors associated with EPS highlight the uncertainty and challenges in identifying clear causative factors. Time on PD is the leading risk factor for EPS with the incidence between 0.7 and 13.6 per 1000 patient-years with patients more than 5 years on PD in some reports.<sup>149</sup> The mortality rate among patients diagnosed with EPS is alarmingly high, ranging from 26% to 56% in the first year of diagnosis and higher mortality rates are seen with longer PD vintage.<sup>147</sup> Nonetheless, recent figures indicate better survival rates, likely due to earlier identification and treatment of EPS.

Despite the lack of universally accepted approaches to EPS treatment, management focuses on a combination of supportive, surgical, and pharmacotherapeutic options that should be individualized to meet patient needs.<sup>147,149,150</sup> Patients typically require nutritional support, and, in some instances, total parenteral nutrition is essential. In terms of surgical interventions, patients can undergo a procedure to release bowel adhesions.<sup>150</sup> This requires highly specialized surgical skills to prevent complications such as bowel perforation. Medical treatments can be either supportive or aimed at mitigating the inflammation and scarring seen in EPS. Tamoxifen, an antifibrotic drug, is seen as a potential therapeutic option for EPS.<sup>154</sup> Observational data have shown some benefit with sirolimus and glucocorticoids in the management of EPS as well.<sup>151,154</sup> Presently, prevention remains elusive due to limited understanding of EPS's origins. However, early detection and intervention could prove critical in managing this condition effectively. Ultimately, in managing long-term dialysis (>5 years on PD), providers should assess the risks of EPS against HD-associated risks

including infection of the vascular access and vascular access-related complications, along with additional dietary and other restrictions affecting lifestyle.<sup>149</sup> For younger patients on PD for extended periods, providers should discuss the possibility of transitioning to home HD or pursuing transplantation. For older patients, the decision to switch modalities must consider the impact on quality of life versus the likelihood of competing mortality risk from comorbid conditions.<sup>155</sup> In cases where patients have a prolonged PD duration with an increased EPS risk and low mortality risk, providers may consider HD after an informed discussion with the patient. It is also to recognize that EPS most commonly presents in cases of PD withdrawal after many years on treatment.

## INFECTIOUS COMPLICATIONS

### EXIT SITE AND TUNNEL INFECTIONS

An exit site and tunnel infections are PD-catheter-related infections that can lead to peritonitis, hospitalization, HD transfer, and death. Prompt recognition and treatment are important to reduce risk of complications for patients on PD. An exit site infection is defined by the presence of purulent drainage, which may or may not be accompanied by erythema at the catheter-epidermal interface.<sup>156</sup> A tunnel infection is defined by inflammation, which may include erythema, swelling, tenderness, or induration with or without ultrasonographic evidence of fluid collection along the tunnel site of the catheter.<sup>156</sup> The 2023 updates to ISPD guidelines on catheter-related infections represent a significant shift in the management of exit site and tunnel infections.<sup>156</sup> These revisions redefine the criteria for these infections, setting a goal to keep exit site infection rates below 0.40 episodes per year at risk.<sup>156,157</sup> The guidelines have also modified recommendations for the use of topical antibiotic creams or ointments at the catheter exit site, now emphasizing the importance of exit site dressing cover and updated antibiotic treatment durations, with a focus on early clinical monitoring. Diagnosing exit site infections relies primarily on clinical assessment, especially noting changes from a patient's normal healthy exit site.<sup>157</sup>

In terms of treatment, the ISPD guidelines recommend a tailored approach based on the type of infecting organism. For uncomplicated infections, a general recommendation is a 7- to 10-day antibiotic course, extending as needed for more complex conditions like tunnel infections or infections caused by virulent organisms. For infections caused by gram-positive bacteria such as *Staphylococcus aureus*, including methicillin-resistant *S. aureus*, antibiotics like vancomycin or linezolid are recommended.<sup>157</sup> For gram-negative bacterial infections including those caused by *Pseudomonas* species, treatment typically involves antipseudomonal (extended spectrum) penicillins or cephalosporins, potentially combined with aminoglycosides or quinolones. Fungal infections require antifungal agents, such as fluconazole or amphotericin B, depending on the specific fungus.

### PERITONITIS: DEFINITION, DIAGNOSIS, AND CLINICAL COURSE

Peritonitis remains one of the most significant complications of PD and is associated with patient morbidity, mortality, and

cost to the health care system (see Table 63.1 for definitions of peritonitis).<sup>158,159</sup> Given the evolving nature of practice in PD along with microbiological resistance patterns, there have been eight iterations of ISPD recommendations on the prevention and treatment of peritonitis over the past 4 decades with the most recent being published in 2022.<sup>100</sup> In 2022, the ISPD refined definitions for various types of peritonitis and their associated complications. These include refractory, relapsing, and peritonitis leading to catheter removal, HD transfer, hospitalization, and death.

**Table 63.1 2022 ISPD Definitions of Peritonitis-Related Outcomes and Various Types of Peritonitis**

| Term                                    | Definition                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Medical cure                            | Complete resolution of peritonitis together with NONE of the following complications: relapse/recurrent peritonitis, catheter removal, transfer to HD for $\geq 30$ days or death                                                                                                                                        |
| Refractory                              | Peritonitis episode with persistently cloudy bags or persistent dialysis effluent leukocyte count $>100 \times 10^9/L$ after 5 days of appropriate antibiotic therapy                                                                                                                                                    |
| Recurrent                               | Peritonitis episode that occurs within 4 weeks of completion of therapy of a prior episode but with a different organism                                                                                                                                                                                                 |
| Relapsing                               | Peritonitis episode that occurs within 4 weeks of completion of therapy of a prior episode with the same organism or one sterile (culture negative) episode (i.e., specific organism followed by the same organism, culture negative followed by a specific organism or specific organism followed by culture negative). |
| Repeat                                  | Peritonitis episode that occurs more than 4 weeks after completion of therapy of a prior episode with the same organism                                                                                                                                                                                                  |
| Peritonitis-associated catheter removal | Removal of PD catheter as part of the treatment of an active peritonitis episode                                                                                                                                                                                                                                         |
| Peritonitis-associated HD transfer      | Transfer from PD to HD for any period as part of the treatment for a peritonitis episode                                                                                                                                                                                                                                 |
| Peritonitis-associated death            | Death occurring within 30 days of peritonitis onset or death during hospitalization due to peritonitis                                                                                                                                                                                                                   |
| Peritonitis-associated hospitalization  | Hospitalization precipitated by the occurrence of peritonitis for the purpose of peritonitis treatment delivery                                                                                                                                                                                                          |

HD, Hemodialysis; ISPD, International Society for Peritoneal Dialysis; PD, peritoneal dialysis.

From Li PK, Chow KM, Cho Y, et al. ISPD peritonitis guideline recommendations: 2022 update on prevention and treatment. *Perit Dial Int*. 2022;42(2):110–153.

Additionally, new categories for cause-specific peritonitis were introduced including pre-PD, enteric, and catheter-related peritonitis.<sup>100</sup> Despite these changes in guidelines, the overarching definition of peritonitis has remained unchanged. The diagnosis of peritonitis is made when at least two of the following criteria have been met: 1. clinical features such as abdominal pain and/or cloudy effluent; 2. dialysis effluent white cell count  $>100/\mu L$  or  $>0.1 \times 10^9/L$  (after a dwell time of at least 2 hours), with  $>50\%$  polymorphonuclear leukocytes; and 3. positive dialysis effluent culture.<sup>100</sup> A reliable diagnosis of peritonitis is based on efficient cultures of the peritoneal fluid. When done correctly, ideally more than 80% should be able to culture a microorganism. The preferred method involves placing 5 to 10 mL of this fluid directly into blood culture containers.<sup>160,161</sup>

The earliest clinical manifestation often presents as a turbid peritoneal effluent, though some patients may present with abdominal pain before changes in effluent appearance. Although cloudy effluent is a hallmark of peritonitis, differential diagnoses are broad. Potential causes of cloudy effluent include fibrin, chylous ascites, malignancy, eosinophilic peritonitis, and instances of dry peritoneum sampling. It is worth noting that eosinophilic peritonitis is characterized by an absolute number of eosinophils  $>40$  cells, corresponding to more than 10% of total white blood cells in a turbid effluent.<sup>162</sup> The cause of eosinophilic peritonitis is thought to be connected to allergic reactions to specific components of the PD system, leading to hypersensitive responses. It often manifests shortly after the insertion of the PD catheter or subsequent alterations to the PD setup. Possible triggers range from the PD catheter itself, certain medications, the dialysate solution, or occasionally, mycotic infections.<sup>162,163</sup>

Bacteria can enter the peritoneal cavity by two primary methods, through the catheter (intraluminal route) and along the catheter (pericatheter route). Additionally, there is now an increased awareness of bacteria moving across the bowel mucosa or other tissue layers (transmural route). This is often linked to gastrointestinal conditions such as abscesses, diverticulitis, or constipation, but more often enteric organisms may enter the peritoneum via transmural bowel migration even in the absence of overt bowel compromise. Furthermore, the nasal passages and skin being colonized by *S. aureus* are known to be associated with infections where the catheter exits the body and other catheter-linked issues, both significant contributors to peritonitis. Additional risk factors for peritonitis encompass touch contamination, bacterial or yeast infections originating from transvaginal routes and systemic bacteremia.<sup>100,164,165</sup>

## GRAM-POSITIVE PERITONEAL DIALYSIS PERITONITIS

Analysis of PDOPPS data from seven countries showed that gram-positive bacteria remain the leading cause of peritonitis, particularly with coagulase-negative *Staphylococcus* and *S. epidermidis*.<sup>166</sup> These organisms are likely from touch contamination and colonization from skin flora. The type of pathogens can differ widely on the basis of skin flora and local microbiological patterns. The next most common bacterial culprit of PD peritonitis is

*S. aureus*, often linked with tunnel or exit site infections. When compared with *S. epidermidis*, *S. aureus* peritonitis typically results in more severe illness, increasing the risk of catheter loss and a potential switch to hemodialysis, either temporarily or permanently. Enterococci rank third among the agents causing gram-positive peritonitis. While they generally respond to antimicrobial treatment, it's essential to check for bacterial sensitivity due to the prevalence of vancomycin-resistant strains. Issues like constipation and other bowel-related problems can increase susceptibility to enterococcal infections, which often lead to repeated and persistent peritonitis. (See Table 63.2 for treatment duration and other important considerations in management of other gram-positive organism-related PD peritonitis.)

### GRAM-NEGATIVE PERITONEAL DIALYSIS PERITONITIS

While gram-positive organisms remain the leading cause of PD peritonitis, the relative incidence of gram-negative infections has risen over the past 2 decades.<sup>167,168</sup> These infections can stem from various sources including the bowel, skin, urinary tract, contaminated water sources, and even with animal contact. Infections from *Escherichia coli* and other Enterobacteriaceae like *Klebsiella* or *Proteus* are especially concerning in developing countries but generally respond to antibiotics.<sup>169,170</sup> Additionally, nonfermenting gram-negative bacilli play a crucial role as nosocomial pathogens causing severe peritonitis. These include *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, and *Stenotrophomonas maltophilia*.<sup>100</sup> *Pseudomonas* peritonitis accounts for about 8% to 10% of gram-negative peritonitis cases, which can be severe. It is also associated with a <50% cure rate, often requiring catheter removal, which is partly explained by the development of biofilms on the catheter with pseudomonal PD peritonitis.<sup>100,166,167,171</sup> Improved outcomes are observed with early catheter removal, temporary transition to hemodialysis, and dual-antipseudomonal antibiotic treatment, allowing many

patients to resume PD. Although historically common, double antibiotic coverage for pseudomonal infections is no longer recommended by clinical guidelines for managing PD-associated infections in children, based on emerging evidence.<sup>172</sup>

Practitioners should be aware of organisms like extended-spectrum  $\beta$ -lactamases (ESBL) and “SPICE” organisms, primarily *Serratia*, *Providencia*, indole-positive *Proteus*, *Citrobacter freundii*, and *Enterobacter* species, which are increasingly relevant in peritonitis cases. ESBL, a diverse group of mainly plasmid-mediated enzymes, deactivate  $\beta$ -lactam antibiotics and are linked to poorer clinical outcomes.<sup>173</sup> These ESBL producers, often resistant to fluoroquinolones and aminoglycosides, have shown a notable increase in causing peritonitis, with treatment failures correlating with resistance to second- and third-generation cephalosporins along with fluoroquinolones.<sup>174</sup> In such cases, prompt PD catheter removal may be necessary. AmpC enzymes, primarily produced by ‘SPICE’ organisms, can induce resistance to earlier-generation cephalosporins, even when in vitro tests show susceptibility. Treatment options should consider fourth-generation cephalosporin, quinolones, or carbapenems.<sup>100</sup> (See Table 63.2 for treatment duration and other important considerations in management of other gram-negative organisms and polymicrobial PD peritonitis).

### FUNGAL AND MYCOBACTERIAL PERITONEAL DIALYSIS PERITONITIS

Fungal peritonitis is a less common cause of peritonitis but can lead to poor outcomes for patients. A PDOPPS review has highlighted an increased risk of death, transfer to HD, and catheter removal when compared with gram-positive PD peritonitis.<sup>166</sup> Observational data have shown that mortality rate associated with fungal PD peritonitis is high and may be attenuated with early catheter removal.<sup>175</sup> *Candida albicans* is the most common species involved in fungal peritonitis, and its presence alongside bowel obstruction and abdominal

**Table 63.2 Recommended Duration of Antimicrobial Treatment for Various Organisms Associated With Peritoneal Dialysis Peritonitis and Important Considerations**

| Organism                                         | Treatment Duration | Important Considerations                                                                                                                                                                                           |
|--------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CNST (Coagulase-negative <i>Staphylococcus</i> ) | 2 weeks            | <ul style="list-style-type: none"> <li>Switch to vancomycin if methicillin resistant even if cell count improving on a cephalosporin</li> </ul>                                                                    |
| <i>Staphylococcus aureus</i>                     | 3 weeks            | <ul style="list-style-type: none"> <li>Consider adjuvant rifampin</li> </ul>                                                                                                                                       |
| Streptococci                                     | 2 weeks            | <ul style="list-style-type: none"> <li>Check oral flora and dental hygiene</li> </ul>                                                                                                                              |
| Enterococci                                      | 3 weeks            | <ul style="list-style-type: none"> <li>Consider CAPD if ampicillin sensitive in every exchange</li> </ul>                                                                                                          |
| Nonpseudomonal gram negative                     | 3 weeks            | <ul style="list-style-type: none"> <li>SPICE/ESBL consider longer treatment time, 2 antibiotics to reduce relapse risk</li> </ul>                                                                                  |
| <i>Corynebacterium diphtheroid</i>               | 3 weeks            | <ul style="list-style-type: none"> <li>Laboratories may not report sensitivities (may treat as contaminant)</li> <li>Switch to vancomycin</li> </ul>                                                               |
| <i>Pseudomonas</i>                               | 3 weeks            | <ul style="list-style-type: none"> <li>Close examination of exit and tunnel site</li> </ul>                                                                                                                        |
| <i>Stenotrophomonas</i>                          |                    | <ul style="list-style-type: none"> <li>Use Sulfamethoxazole-Trimethoprim if sensitive to <i>Stenotrophomonas</i></li> </ul>                                                                                        |
| Culture negative                                 | 2 weeks            | <ul style="list-style-type: none"> <li>If cell count improving, can stop gram-negative coverage</li> <li>If cell count not improving, consider unusual organisms (i.e., typical, atypical mycobacteria)</li> </ul> |
| Polymicrobial                                    | 3 weeks            | <ul style="list-style-type: none"> <li>If multiple enteric organisms, imaging to rule out surgical peritonitis</li> <li>Consider addition of empiric anaerobic coverage</li> </ul>                                 |



pain may portend a poor outcome for patients. Antibacterial treatments can enhance the risk of fungal peritonitis by suppressing normal bowel flora, leading to fungal growth and transmigration to the peritoneum.

Mycobacterial peritonitis is rare, often initially diagnosed as culture-negative PD peritonitis. The ISPD recommends requesting for Ziehl-Neelsen staining for acid-fast bacilli in cases where there is a high clinical suspicion of nontuberculous mycobacterial peritonitis, particularly in instances of continuous culture-negative peritonitis.<sup>100</sup> While acid-fast smears are not always reliable, PCR techniques are effective in detecting tuberculous genes in peritoneal fluid; if available in local centers, polymerase chain reaction testing should be considered.<sup>176</sup> Additionally, in tuberculosis-endemic regions, it may account for up to 13% of peritonitis cases, posing significant risks, especially in malnourished and immunocompromised patients.<sup>177,178</sup> It should be noted that atypical mycobacteria peritonitis is usually from a PD catheter source, whereas tuberculous mycobacteria may be due to systemic infection resulting in spreading to the peritoneum.

### TREATMENT AND PREVENTION OF PERITONEAL DIALYSIS PERITONITIS

Following preliminary diagnostic evaluations for suspected PD peritonitis, immediate initiation of intraperitoneal (IP) antibiotics with a minimum dwell of 6 hours is imperative for optimal pharmacokinetics.<sup>100,179</sup> The IP administration ensures increased and appropriate antibiotic concentrations at the infection site. The more common practice involves administering antibiotics once a day, either through a daytime dwell in CAPD or a manual exchange in APD. It is essential to choose antibiotics that cover both gram-negative (including antipseudomonal) and gram-positive organisms, considering local resistance trends. Adjustments should then be made on the basis of culture and sensitivity results. Registry data have revealed that patients receiving broad empirical antibiotics at initial presentation had a higher likelihood of achieving PD peritonitis cure compared with delayed treatment.<sup>166,180</sup> Typically, signs like cloudy effluent diminish within 48 to 72 hours of antibiotic administration. If symptoms persist, it may be suggestive of resistant bacterial strains, warranting either an alteration in the antibiotic regimen or catheter removal.<sup>168</sup> As a result, monitoring effluent cell counts within the initial 5 days is instrumental in evaluating treatment effectiveness. Current guidelines advise a combination of either cephalosporin or vancomycin for gram-positive bacteria and a third-generation cephalosporin or an aminoglycoside for gram-negative ones.<sup>100</sup> Though vancomycin is efficacious against gram-positive bacteria, there have been some concerns about resistance. However, resistance is rare and has not truly emerged as a prevalent problem in clinical practice. (See [Table 63.2](#) for treatment duration for various organisms.)

Proper PD catheter placement, allowing for clear exit site visualization and care, is crucial for peritonitis prevention (see “[Preparing for Peritoneal Dialysis Catheter Insertion](#)”). Daily mupirocin application at the exit site can combat *S. aureus* infections, though its optimal frequency remains undefined despite strong evidence supporting its efficacy.<sup>181</sup> Local gentamicin use has proven effective against *Pseudomonas* and other gram-negative infections.<sup>47</sup> A small observational study demonstrated gentamicin superiority compared with

mupirocin in preventing exit site infections;<sup>182</sup> however, there is emerging concern about increased resistance with continued gentamicin use. It is essential for patients to recognize and report a “wet contamination” to the dialysis facility, with responses possibly including changing transfer sets and/or using prophylactic antibiotics. Wet contamination involves exposure to contaminants via leaks in dialysate bags, tubes, or connections during PD fluid handling or through breaches in tubing near the clamp.<sup>165</sup> It can also include direct contact contamination at the connection point during PD exchanges. Employing double-bag systems in ambulatory patients and a flush-before-fill approach in automated setups has also notably diminished the incidence of peritonitis.

Peritonitis in PD patients often follows endoscopic gastrointestinal and gynecologic procedures like gastroscopy and hysteroscopy, with the highest rates observed after invasive gynecologic procedures (26.9%–38.5%).<sup>100,183</sup> Rates after colonoscopy without antibiotics range from 3.4% to 8.5%, and gastroscopy rates range from 1.2% to 3.9%.<sup>100,184–186</sup> Concerns arise due to the pelvic proximity to the peritoneal cavity, with *Streptococcus* and *E. coli* being common pathogens. Studies suggest antibiotic prophylaxis can significantly reduce peritonitis incidence, but there is no standardized recommendation for antibiotic choice.<sup>186</sup> For colonoscopy, more than half of peritonitis cases are caused by *E. coli* and studies show that antibiotic prophylaxis can effectively prevent peritonitis. However, the optimal antibiotic regimen remains undetermined but should have gram-positive and gram-negative coverage. Additionally, draining PD effluent before these procedures is also advised to reduce the risk of infection.<sup>100</sup>

## PRESCRIBING PERITONEAL DIALYSIS

### GOAL-DIRECTED PERITONEAL DIALYSIS

Historically, PD management focused on small solute kinetics with an overemphasis on weekly standardized urea removal targets of a total  $Kt/V_{urea}$  2.0 per week (renal and peritoneal). However, numerous studies have failed to show any significant clinical benefit with values higher than a weekly total  $Kt/V$  of 1.7.<sup>187–189</sup> Evidence supports that the preservation of residual kidney function rather than adhering strictly to standardized weekly clearance targets is associated with better clinical outcomes for patients on PD.<sup>188</sup> The ADEMEX trial was a landmark randomized controlled study and also demonstrated that peritoneal clearance had minimal impact on patient survival.<sup>190</sup> The importance of residual kidney function in this population extends far beyond mere numbers, as it is linked to a substantially lower relative risk of death. In the reanalysis of the multicenter observational CANUSA study, data revealed that a 250 mL incremental increase in urine volume was associated with a 36% lower relative risk of death in patients receiving PD. Additionally, every 5 L/week/1.73 m<sup>2</sup> higher renal clearance was associated with a 12% reduction in mortality risk.<sup>188</sup>

Given the emerging evidence, particularly around the importance of preservation of residual kidney function rather than achieving  $Kt/V_{urea}$  targets, there has been a refocus on tailoring treatment that addresses patient-specific needs. In 2018, the Kidney Disease Improving Global Outcomes

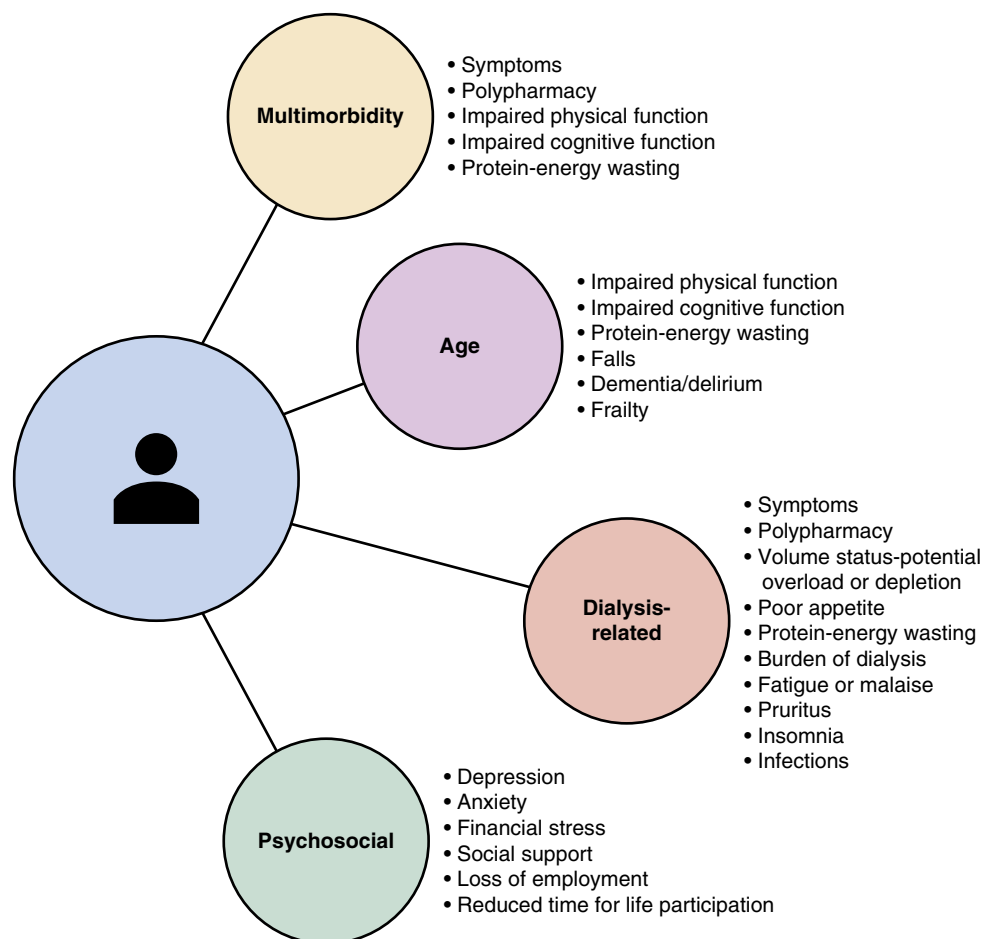


Controversies Conference emphasized transitioning from “adequate” to “goal-directed” dialysis.<sup>191</sup> This approach, based on shared decision making, prioritizes personalized care and considers multiple factors like symptoms, kidney function, and patient well-being to assess dialysis quality. Additionally, practice recommendations from the ISPD also support a treatment approach that promotes goal-directed therapy for patients receiving PD. The main objective of goal-directed PD is to formulate achievable care goals that accomplish three main outcomes.<sup>62</sup> First and foremost, the focus should be on sustaining the quality of life of the individual undergoing PD. This involves ensuring they can pursue their personal aspirations and achieve life goals. The second aim is to reduce the symptoms and lessen the demands of the treatment on the patient. Lastly, the care provided must adhere to the highest standards of quality. As a result, practitioners should focus on addressing a variety of factors for individual patients that could influence their outcomes on PD (Fig. 63.4).

### INCREMENTAL PERITONEAL DIALYSIS

Incremental PD has been recognized as a practice that closely aligns with goal-directed therapy as outlined by the ISPD in 2020.<sup>62</sup> Incremental PD can be described as an initial prescription that includes either CAPD with fewer

than four daily dwells, dwell volumes under 2 L, or fewer than 7 days per week or automated PD without a day dwell, total daily doses below 10 L, or fewer than 7 days per week of treatment<sup>57-59,192,193</sup> (Fig. 63.5). Importantly, incremental PD starts simultaneously with “full-dose” PD and is not a method that advocates for early initiation of dialysis. By offering patients a more tailored approach, incremental PD ensures that therapy is introduced in a manner that gradually increases in intensity as the residual kidney function declines.<sup>57-59,194</sup> This approach not only allows patients to familiarize themselves with the treatment but also maintains a balance between sustaining residual kidney function and ensuring adequate solute clearance. Longitudinal data indicate that when compared with standard PD prescriptions, patients on incremental PD have comparable survival rates but benefit from reduced hospitalizations and costs.<sup>195-197</sup> Furthermore, incremental PD shifts the focus from solely relying on  $Kt/V_{urea}$  to a more comprehensive evaluation involving clinical assessments and other biochemical data. Despite the associated benefits of incremental PD, it is important to address certain challenges, such as therapeutic inertia, in which patients may be hesitant to adjust PD prescriptions in the presence of a decline in residual kidney function.<sup>193,198</sup> Therefore educating both patients and caregivers about incremental PD becomes paramount to facilitate timely changes in the prescription when needed.



**Fig. 63.4** Factors influencing outcomes for patients on peritoneal dialysis when using an individualized approach to care. (Adapted with modification from Brimble KS, Walker M, Margetts PJ, et al. Meta-analysis: peritoneal membrane transport, mortality, and technique failure in peritoneal dialysis. *J Am Soc Nephrol*. 2006;17(9):2591–2598.<sup>53</sup>)

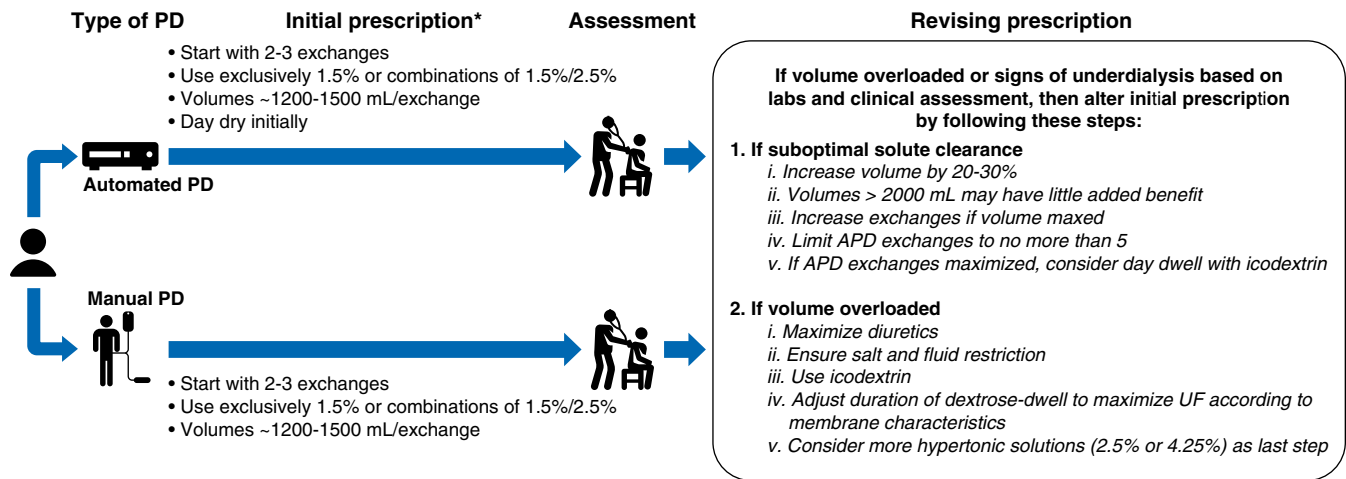


Fig. 63.5 Oversimplified approach of an incremental peritoneal dialysis prescriptive strategy. (© Bourne L. Auguste.)

Incorporating incremental PD means that clinicians must devise prescriptions based on individual patient needs, considering clinical symptoms alongside laboratory data. For those preferring manual exchange therapy, starting with CAPD involving 1 to 2 daily exchanges of 1.5 L each can be an ideal introduction.<sup>192</sup> As the patient grows more accustomed, fill volumes can be gradually increased, with additional daytime exchanges introduced if further solute clearance becomes necessary. Conversely, for those inclined toward cycler-based therapy, nocturnal intermittent peritoneal dialysis (NIPD) can serve as an introductory method, especially benefiting patients with limited daytime flexibility. Beginning with 2 to 3 overnight exchanges, each with 1.5 liters, adjustments in fill volumes or additional exchanges can be made on the basis of residual kidney function decline. Patients can also be offered 1 to 2 days off per week with both manual and automated PD prescriptions as part of incremental PD prescriptions.<sup>58,59</sup> Regardless of the approach, it is vital to monitor patients on incremental PD more meticulously, especially initially, to avoid potential complications of underdialysis, particularly when there is a rapid decline in residual kidney function.

## PERITONEAL DIALYSIS IN FRAIL, MULTIMORBID, OR OLDER PATIENTS

Older patients form a distinctive subset within the PD population, necessitating specialized care and considerations. The integration of assisted PD with goal-directed therapy represents the future of PD care for older patients. Assisted PD, in which trained health care professionals facilitate dialysis exchanges, has increased access to PD for frail and older patients.<sup>199</sup> This convergence offers a comprehensive approach, assuring elderly patients of both a tailored therapeutic regimen and consistent support, leading to improved patient outcomes.<sup>200</sup> Assisted PD is also associated with reduced costs when compared with IHD and can lead to increased uptake of home-based treatments in historically populations deemed ineligible for PD therapy.<sup>201</sup> Additionally, an ISPD position statement outlines key aspects of assisted PD including justifications, implementation models, and requirements for training and

quality control. It recommends health care systems fund assisted PD to increase home dialysis access, supporting both short-term and long-term assistance.<sup>202</sup>

However, assisted PD adoption varies worldwide, largely influenced by government reimbursement policies. France is a notable leader in Europe when it comes to assisted PD with almost half of patients on PD benefiting from assisted programs.<sup>203,204</sup> Twenty years ago, France initiated the assisted PD program, initially targeting only focusing on patients receiving CAPD, but as the years passed, this initiative expanded to include patients on APD. In subsequent years, other European countries have adopted assisted PD programs.<sup>204,205</sup> Reimbursement for assisted PD by the national public health services exists in five countries: Denmark, France, Norway, Sweden, and the United Kingdom. However, the utilization of assisted PD differs among centers within these countries. In Europe, the United Kingdom follows France in having the second-largest assisted PD program, with a prevalence ranging from 5% to 20% across its different centers.<sup>205</sup> It is noteworthy that while research suggests comparable mortality and hospitalization rates between traditional methods and assisted PD, these findings stem from observational analyses.<sup>206</sup> There is also an indication that nursing support can potentially diminish premature HD transitions among patients.<sup>201,207</sup> Additionally, care models that use nurses compared with those employing nonnursing assistance can be tailored to fit the specific resources, needs, and regulatory standards of the local environment. Overall, assisted PD presents a viable alternative for older individuals and others who face challenges in self-managing PD, proving to be an economical option that can foster additional PD uptake.

## CONCLUSION

PD has advanced significantly, with the PDOPPS and SONG-PD initiatives contributing to its optimization and standardization. These efforts ensure the continued efficacy of PD aligns with patient-centered outcomes, bolstering its global adoption. In addition, the ISPD continues to play a pivotal role, educating practitioners on best practices to

minimize complications and the risk of transitioning to HD prematurely. Such informed guidance promises to nurture the sustainable growth of PD, affirming its position as a critical component of kidney replacement therapy. With these concerted efforts, the outlook for expansion and improvement of PD remains promising, ensuring it remains a reliable and accessible treatment for kidney patients worldwide.

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## QUESTIONS

1. A 65-year-old man on automated peritoneal dialysis for the past 6 months presents with increasing swelling in the genital area over the past week. He denies fever, abdominal pain, or cloudy effluent. History is notable for recent introduction of a last (daytime) fill of icodextrin. On physical examination, there is clear fluid accumulation in the scrotum, suggesting genital edema. His PD exchanges continue to drain adequately, but there is noticeable discomfort. What is the most appropriate next step in managing this complication?

- Increase the dwell volume to improve solute clearance.
- Switch to automated peritoneal dialysis (APD) with lower fill volumes.
- Perform a CT peritoneogram to evaluate for a peritoneal leak.
- Transition to hemodialysis due to complication.

**Answer:** c

**Rationale:** The presence of scrotal or genital edema in this case is likely due to an inguinal hernia from a patent processus vaginalis. The recent addition of a day exchange likely contributed to increased intraperitoneal pressure exacerbating the movement of fluid via the patent processus vaginalis. The most appropriate diagnostic step is to perform a CT peritoneogram to confirm the peritoneal leak and assess the extent of the hernia. Once confirmed, surgical repair of the hernia or patent processus vaginalis may be necessary, and adjustments to the PD regimen (such as temporarily reducing dwell volumes or supine PD) may help manage the leak. Transitioning to hemodialysis is only necessary if conservative measures fail or the leak persists.

2. A 70-year-old woman on peritoneal dialysis presents to the emergency department with severe abdominal pain and cloudy dialysate. The team sends her peritoneal effluent for cell count and culture but delays starting antibiotics. Eight hours later, the results reveal 4000 WBCs/ $\mu$ L with 85% neutrophils, and Gram stain shows gram-positive cocci in pairs. Antibiotics are then started. What is the leading cause of decreased PD survival in this case?

- Delay in initiating antibiotics
- Presence of gram-positive cocci
- High WBC count
- Abdominal pain at presentation

**Answer:** a

**Rationale:** Delays in administering antibiotics in cases of PD-related peritonitis significantly reduce PD survival. Observational data indicate that every hour of delay in antibiotic initiation is associated with a 5% to 6% decrease in PD survival. Early intervention is critical to improving outcomes. The presence of gram-positive cocci and elevated WBCs confirms peritonitis, but the delay in starting antibiotics is the most significant factor influencing survival.

3. A 58-year-old woman on PD presents with purulent drainage and erythema at her catheter exit site, accompanied by mild discomfort along the subcutaneous tunnel. She is diagnosed with both an exit site infection and a tunnel

infection. The care team initiates broad-spectrum antibiotics and obtains cultures. Upon receiving the culture results, what is the most appropriate next step in management?

- Narrow antibiotic coverage to target gram-positive bacteria.
- Remove the PD catheter immediately.
- Continue broad-spectrum antibiotics regardless of culture results.
- Switch to intraperitoneal antibiotics.

**Answer:** a

**Rationale:** In managing exit site and tunnel infections in patients on peritoneal dialysis, narrowing antibiotic coverage based on culture results is the most appropriate next step. While broad-spectrum antibiotics are initially necessary due to the potential severity of the infection, targeted therapy is crucial once the causative organism is identified. This approach optimizes treatment effectiveness, reduces the risk of antibiotic resistance, and minimizes unnecessary drug exposure. Typically, these infections are caused by gram-positive bacteria such as *Staphylococcus aureus*. Immediate catheter removal is overly aggressive at this stage and reserved for cases unresponsive to antibiotics or if peritonitis develops. Continuing broad-spectrum antibiotics regardless of culture results is inappropriate as it increases the risk of resistance and side effects. Switching to intraperitoneal antibiotics is not the standard approach for exit site and tunnel infections, as this route is primarily used for treating peritonitis. By tailoring the antibiotic regimen to the specific pathogen, clinicians can provide effective treatment while adhering to principles of antibiotic stewardship, ultimately promoting better outcomes for the patient.

4. A nephrologist is reviewing the case of a 45-year-old woman who has been on peritoneal dialysis for 2 years due to end-stage kidney disease secondary to diabetic nephropathy. The patient has maintained some residual kidney function since starting peritoneal dialysis; however, her latest assessments show a decline in residual urine output. Concerned about the implications of this decline, the nephrologist discusses the case with nephrology fellows, asking about the significance of residual kidney function in patients on dialysis. Which of the following statements about residual kidney function is most accurate in this patient's case?

- Residual kidney function is a predictor of survival in patients on PD but not in patients on HD.
- The contribution of residual kidney function to total small solute clearance primarily explains the survival advantage conferred by its preservation.
- Residual GFR (calculated as the average of urea and creatinine clearances) but not residual urine volume is a predictor of survival in patients on peritoneal dialysis.
- Loss of residual kidney function is associated with a sustained decrease in middle molecule clearance, such as  $\beta$ -2 microglobulin, which is not fully compensated by PD alone.

**Answer:** d

**Rationale:** The decline in residual kidney function in patients on peritoneal dialysis is associated with a significant and sustained decrease in middle molecule clearance, such as  $\beta$ -2 microglobulin. Middle molecules are larger than small solutes, and while PD effectively clears small solutes like urea and creatinine, it is less efficient at clearing middle molecules, which can accumulate when residual kidney function diminishes. This reduced clearance has been linked to adverse outcomes including increased cardiovascular risk and mortality. Option A is incorrect because residual kidney function is a predictor of survival in patients on both PD and HD. Option B is also incorrect, as the survival benefit of residual kidney function extends beyond small solute clearance to include fluid management, phosphorus clearance, and removal of larger molecules. Option C is incorrect because both residual GFR and residual urine volume independently predict survival in patients on PD. The preservation of urine output aids in volume control and reduces fluid overload, which is crucial for long-term survival. The focus on middle molecule clearance in option D highlights the broader implications of residual kidney function beyond small solute clearance, underscoring its role in reducing the risk of accumulating uremic toxins that peritoneal dialysis alone may not fully manage.

5. A 40-year-old female with ESKD has been on automated peritoneal dialysis for the past year. Her total Kt/Vurea is 1.7, and she produces 500 mL of urine daily. She feels well, reports no symptoms, and is adherent to her prescribed therapy. However, the treatment team is concerned about her dialysis adequacy. What is the most appropriate next step?
  - a. Increase Kt/Vurea to 2.0.
  - b. Continue current regimen.
  - c. Switch to continuous ambulatory peritoneal dialysis.
  - d. Start hemodialysis.

**Answer:** b

**Rationale:** This patient is doing well on her current regimen with stable residual kidney function. Goal-directed PD as outlined by the International Society of Peritoneal Dialysis focuses on shared decision making and tailoring therapy to the patient's well-being and life goals rather than strictly focusing on achieving higher Kt/Vurea targets. Increasing Kt/Vurea is not necessary if the patient is asymptomatic, and switching modalities to continuous ambulatory peritoneal dialysis or starting hemodialysis would increase the treatment burden without added benefit. Continuing the current regimen is appropriate, with regular monitoring of symptoms and for any changes in residual kidney function.